



AIR HOISTS



CoreLifting.com





AIR HOISTS

JDN BBH SERIES

Robust – reliable – low maintenance
The air hoist series for big bag handling

BBH 1000 AND BBH 2000

All JDN air hoists in the BBH series are designed and manufactured to be extremely robust. From our ongoing dialogue with our customers, we know that safety and reliability are top priorities wherever work is carried out on a sustained basis and under demanding conditions. This is precisely why the JDN air hoists in the BBH series are highly valued and successfully used in these cases.

The hoists in the BBH series offer 100% delay-free switch-on and unlimited switch-on frequency. This is ensured by the non-slip, low-maintenance multi-blade motor. Thanks to the pneumatic drive, they offer explosion protection

(ATEX) as standard and are also resistant to dust and moisture. This makes them ideal not only for use in potentially explosive atmospheres, but also for demanding industrial applications and continuous operation – at temperatures ranging from -20°C to +70°C.

JDN BBH hoists are easy to operate for all applications and have a sensitive, continuously variable control system for positioning loads safely and precisely. Various control systems are available for this purpose, depending on requirements.

From a load-carrying capacity of 1 t, the devices in the BBH series are also

equipped with overload protection as standard. Thanks to dust-protected ball bearings, a permanently lubricated planetary gear with high-performance lubricant and gears made of hardened or tempered stainless steel, all JDN air hoists in the BBH series require only minimal maintenance.

We remain in close contact with our users at all times through our 360° service partnerships – and we use this dialogue to continuously develop our products.

**In short: we keep our promise
“Permanent Excellence” for you.**

INNOVATION AND VALUES

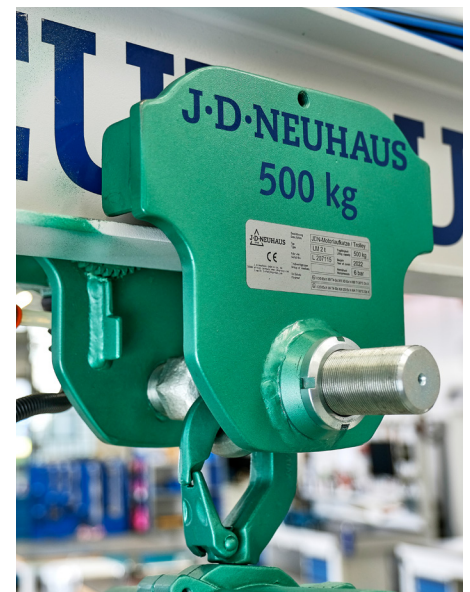
J.D. NEUHAUS – MORE THAN PRODUCTS

As a family-run company in its seventh generation, our commitment to clear values helps us achieve the standard of “Permanent Excellence” that we have held ourselves to for over 275 years.

The three value pairs of tradition and innovation, performance and longevity as well as partnership and passion create the foundation for our daily activities. They shape our partnership with the people

who work for us, as well as with our partners and customers. They are found in the quality and German engineering of our products, and in the way we work with you and find new solutions thanks to our innovative spirit.

Our thinking is guided by sustainability, which we believe our future urgently needs and, of course, deserves.





PRODUCT BENEFITS AND ADVANTAGES IN PRACTICE

With hoists from the BBH series, you as a safety-oriented, high-performance company can expand your equipment portfolio with a reliable tool with power features and benchmark qualities.

JDN BBH hoists are mechanical and are pneumatically operated. Thanks to compact and robust design with few components, the BBH series is perfectly matched to your application environment. As a result, the devices are not susceptible to external influences and are extremely low maintenance.

Our patented pneumatic drive system, designed for 100% power-up time, offers fast lifting and lowering speeds. In this way, we ensure that customers do not experience downtimes and that permanent availability is ensured.

BBH series JDN air hoists ensure maximum performance and safety in any industry or process where they are used. The processed materials and their coating make the hoists insensitive to factors such as paint particles, spray mist, moisture,

aggressive vapours, dirt and dust. They are also resistant to very low or very high temperatures.

Die BBH-Serie is available in two versions. With one load hook for standard cruciform lifting beam designs. The extended distance between the hook and the chain box is particularly advantageous. This guarantees that there is no risk of collision between the load and the chain box. With twin load hooks for more complex cruciform lifting beam designs or for standard lifting beam designs with two suspension points.

Particularly suited for use as big bag handling hoists and for the movement of all kinds of bulky loads due to the low headroom design..

PNEUMATIC
mechanical drive

ROBUST
to external ambient influences

NO DOWNTIME
100% on-time

EXPLOSION SAFE
for use in hazardous areas

COST-EFFECTIVE
thanks to low maintenance

TECHNICAL DATA

TYPE		BBH 1000-1	BBH 2000-1	BBH 1000-2	BBH 2000-2
Capacity	mt	1.1	2.2	1.1	2.2
Air pressure	psi bar	87 6			
Number of hooks		1		2	
Number of chain strands		1	2	2	4
Motor output hoist	kW	0,7			
Motor output trolley	kW	0,2			
Lifting speed at full load	ft/min m/min	12.14 3.7	5.58 1.7	12.14 3.7	5.58 1.7
Lifting speed without load	ft/min m/min	21.33 6.5	11.48 3.5	21.33 6.5	11.48 3.5
Lowering speed at full load	ft/min m/min	21.33 6.5	11.48 3.5	21.33 6.5	11.48 3.5
Air consumption at full load – lifting	cfm m³/min	60.03 1.7			
Air consumption at full load – lowering	cfm m³/min	67.10 1.9			
Air consumption at full load – trolley	cfm m³/min	21.19 0.6			
Air connection		G 1/2			
Hose dimension (Ø inside)	inch mm	1/2 13			
Weight at standard lift and min. k dimension	lbs kg	286.60 130	302.03 137	302.03 137	328.49 149
Chain dimension	mm	7 x 21			
Weight of chain	lbs/ft kg/m	0.67 1			
Standard lift	ft m	10 3			
Length of control at standard load – lift	ft m	6.5 2			
Noise level at full load – lifting	dB(A)	76			
Noise level at full load – lowering	dB(A)	78			
Noise level at full load – trolley	dB(A)	80			

CONTACT

Please contact us for more information. Our colleagues from sales, projectmanagement and service are there for you.

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Singapore, United Kingdom, United States





AIR HOISTS

JDN EXPLOSION PROTECTION

CLASSIFICATION AND MARKING



JDN EXPLOSION PROTECTION CLASSIFICATION AND MARKING

Hoists and cranes from J.D. Neuhaus have an unbeatable advantage over electrically driven lifting equipment: Even the standard versions are suitable for use in explosion hazardous areas.

If you have any questions about the topic of explosion protection, please contact our sales team. We will be happy to advise you.



WHAT EXACTLY DO THE MARKINGS MEAN?

II 3G Ex h IIB T4 Gc X
II 3D Ex h IIIB T130°C Dc X

MARKING FROM ATEX

Zone	Mining	0		1		2	
		Gas	Dust	Gas	Dust	Gas	Dust
Equipment group	I	II					
Equipment category	M1 or M2	1G	1D	2G	2D	3G	3D

IGNITION PROTECTION TYPE

Non-electrical explosion protection is always assigned the marking code "h" irrespective of the version. JDN uses constructive explosion protection.

EXPLOSION GROUP

Mining	Gas		Dust	
-	IIA	E.g. propane	IIIA	Flammable lint
	IIB	E.g. ethylene	IIIB	Non-conductive dust
	IIC	E.g. hydrogen	IIIC	Conductive dust

MAXIMUM SURFACE TEMPERATURE

Mining	Gases		Dusts
-	T1	450°C	T...°C
	T2	300°C	
	T3	200°C	
	T4	135°C	
	T5	100°C	
	T6	85°C	

For areas with a risk of dust explosion, the maximum permissible surface temperature of the device is specified explicitly; for areas with a risk of gas explosion, the maximum surface temperature is divided into a temperature class.

EQUIPMENT PROTECTION LEVEL „EPL“

Zone	Mining	0		1		2	
		Gas	Dust	Gas	Dust	Gas	Dust
EPL	Ma or MB	Ga	Da	Gb	Db	Gc	Dc

ADDITIONAL MARKING

The additional marking "X" refers to particular operating conditions that are



AIR HOISTS

JDN LOW HEADROOM TROLLEYS

Robust – Reliable – Low maintenance
The LMF Series is designed for areas
where headroom is restricted.



◀ 2-TON MODEL

OPTIONAL FEATURES

SPARK-RESISTANT PACKAGES

Including bronze hooks & bottom blocks, stainless steel load chain and hook latches and aluminum, bronze or stainless steel hand chain for geared trolleys.

CORROSION-RESISTANT FINISHES

Created specifically for operation in salt-laden or other corrosive environments.

SWIVEL TROLLEYS FOR CURVED MONORAILS

When using swivel trolleys, each pair of wheels pivots independently of the other to allow the trolley hoist to negotiate tighter curves than a standard rigid trolley. Swivel trolleys also greatly reduce scraping and scrubbing of the wheels against the beam flanges, thus reducing wear.

E-STOP ON CONTROL PENDANT

For added safety.

RUBBER TROLLEY BUMPERS

Helps to absorb the impact of the trolley striking the end stops, protecting the hoist and trolley.

REMOVABLE ANTI-DROP LUGS

As an added safety feature, these easy-to-install anti-drop lugs prevent the trolley from falling off of the monorail in the unlikely event of a wheel or axle failure.

CHAIN CONTAINERS

Chain containers are designed to receive idle load chain as it reeves over the load sheave when lifting a load. This prevents the idle chain from interfering or marring and scratching the load or surrounding surfaces. Compact design helps maintain best possible headroom. Containers are available for most Chester hoist models.

SPECIFICATIONS

Capacity (tons)	Catalog Number			Standard Lift (ft.)	Minimum Headroom (in.)	Minimum Radius Curve **	Hoist Speed (FPM)	CFM at 90 PSI	Weight (lbs.)		
	Plain	Geared	Motorized						Plain	Geared	Motorized
1	ALP-1	ALG-1	ALM-1	10	7	6'6"	10	265	480	510	560
1-1/2	ALP-1 1/2	ALG-1 1/2	ALM-1 1/2	10	7	6'6"	10	265	480	510	560
2	ALP-2	ALG-2	ALM-2	10	7	6'6"	10	265	480	510	560
3	ALP-3	ALG-3	ALM-3	10	8	6'6"	5	265	580	610	660
4	ALP-4	ALG-4	ALM-4	10	8-1/2	7'6"	5	265	580	610	660
5	ALP-5	ALG-5	ALM-5	10	9-1/2	8'6"	7	350	1,640	1,700	1,680
6	ALP-6	ALG-6	ALM-6	10	9-1/2	8'6"	7	350	1,640	1,700	1,680
8	ALP-8	ALG-8	ALM-8	10	12	9'6"	4	350	1,710	1,840	1,770
10	ALP-10	ALG-10	ALM-10	10	12-1/2	10'	3.5	350	2,090	2,170	2,150
12	ALP-12	ALG-12	ALM-12	10	12-1/2	*	3.5	350	2,090	2,170	2,150
16	ALP-16	ALG-16	ALM-16	10	14-1/2	*	2	350	2,660	2,740	2,720
20	ALP-20	ALG-20	ALM-20	10	18	*	1.7	350	3,010	3,170	3,120
24	ALP-24	ALG-24	ALM-24	10	18	*	1.7	350	3,010	3,170	3,120

* Straight track operation only recommended for these units. However curve radius must be specified on orders.

** Curves less than minimum may be fitted by special construction. Please consult factory.

NOTE: All headroom dimensions determined while hoist under load.

THE TROLLEY SOLUTION FOR RESTRICTED HEADROOM AREAS

Where headroom is restricted and standard trolleys can't meet the lifting height requirements we recommend JDN Low Headroom Trolleys whereby our air hoists are mounted horizontally. When only very low headroom is available we recommend JDN Ultra-Low Monorail Hoist design.

STANDARD FEATURES

- Small number of maintenance/ wear free moving parts
- No additional motor lubrication required
- 2-step travelling speed
- Adjustable trolley widths to suit your requirements

SPECIAL FEATURES

- Able to negotiate curves
- Extended trolley tie bars for bulky or elongated loads



Low Headroom Trolley LMF

INNOVATION AND VALUES

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partners and customers. They are found in the quality and German engineering of our products, and in the way we work with you and find new solutions thanks to our innovative spirit.

Our thinking is guided by sustainability, which we believe our future urgently needs and, of course, deserves.



TECHNICAL DATA

HOIST TYPE		PROFI 05 TI	PROFI 1 TI	PROFI 2 TI	PROFI 3 TI	PROFI 6 TI
TROLLEY TYPE		LMF 05-2 T	LMF 05-2 T	LMF 05-2 T	LMF 3.2 T	LMF 6.3 T
Capacity	mt	0.5	1	2	3.2	6.3
Air pressure	psi	87	87	87	87	87
	bar	6	6	6	6	6
Number of chain strands		1	1	2	1	2
Motor output Hoist	kW	1	1	1	3.5	3.5
Motor output Trolley	kW	0.2	0.2	0.2	0.2	0.2
Lifting speed at full load	ft/min	32.81	16.40	8.20	14.76	7.21
	m/min	10	5	2.5	4.5	2.2
Lifting speed without load	ft/min	55.77	32.81	16.40	29.52	14.76
	m/min	17	10	5	9	4.5
Lowering speed at full load	ft/min	55.77	36.09	18.04	35.43	17.72
	m/min	17	11	5.5	10.8	5.4
Travelling speed at full load	ft/min	29.53*/45.93	29.53*/45.93	29.53*/45.93	29.53*/45.93	29.53*/45.93
	m/min	9*/14	9*/14	9*/14	9*/14	9*/14
Air consumption at full load – lifting	cfm	42.38	42.38	42.38	141.26	1141.26
	m³/min	1.2	1.2	1.2	4	4
Air consumption at full load – lowering	cfm	52.97	52.97	52.97	194.23	194.23
	m³/min	1.5	1.5	1.5	5.5	5.5
Air consumption trolley motor	cfm	21.19	21.19	21.19	21.19	21.19
	m³/min	0.6	0.6	0.6	0.6	0.6
Luftanschluss		G ½	G ½	G ½	G ½	G ½
Hose dimension (Ø inside)	mm	13	13	13	19	19
Weight with standard lift and control	lbs	216.5	218.26	231.59	462.97	727.53
	kg	98	99	105	210	330
Chain dimension	inch	0.28 x 0.83	0.28 x 0.83	0.28 x 0.83	0.51 x 1.42	0.51 x 1.42
	mm	7 x 21	7 x 21	7 x 21	13 x 36	13 x 36
Weight of chain	lbs/ft	0.67	0.67	0.67	2.6	2.6
	kg	1	1	1	3,8	3,8
Standard lift	ft	10	10	10	10	10
	m	3	3	3	3	3
Length of control at standard lift	ft	6.5	6.5	6.5	6.5	6.5
	m	2	2	2	2	2
Max. bottom flange thickness t	inch	0.98	0.98	0.98	1.38	1.38
	mm	25	25	25	35	35
Max. bottom flange width b	inch	12.20	12.20	12.20	12.20	12.20
	mm	310	310	310	310	310
Min. bottom flange width b	inch	3.15	3.15	3.15	4.92	4.92
	mm	80	80	80	125	125
Noise level at full load 1 – lifting	dB(A)	75	76	76	78	78
Noise level at full load 1 – lowering	dB(A)	78	78	78	80	80

*1st step at F-control with 2-step travelling speed,

‡Measured at 1 m distance acc. to DIN 45635 part 20



AIR HOISTS

JDN PROFI SERIES

Robust – Reliable – Versatile

CUSTOM MADE FOR YOUR APPLICATIONS

Introducing the JDN PROFI series air hoist — where unparalleled performance and unbeatable reliability meet maximum safety. Engineered to thrive in the toughest industrial environments, our PROFI series is your ultimate solution for various applications.

Imagine a hoist configuration **special-ly** crafted for your application, based on feedback from operators around the world. This lifting solution offers a 100% duty cycle thanks to its non-slip, low-maintenance pneumatic motor. Designed for ATEX environments, the PROFI series excels in potentially explosive atmospheres and dust-filled areas. Depending on your needs, you can customize your hoist with a variety of options, including chains, hooks, chain boxes, and trolleys. And, with its robust, proven design it can withstand toughest conditions making it the top hoist solution for safe lifting.

Our patented pneumatic drive system

guarantees fast lifting and lowering speeds, reducing downtime and keeping your operations running smoothly. Specially lubricated motor technology ensures long-term, oil-free operation, contributing to the JDN sustainability promise. Dust-protected ball bearings and a permanently lubricated planetary gear system mean minimal maintenance and maximum uptime. Additionally, our soundproof motor design ensures a quieter and safer work environment for your staff.

If required the PROFI series features a sensitive, continuously variable control system for precise load positioning. With standard overload protection on models with a load-carrying capacity of 1 ton or more, you can trust in the safety and reliability of every lift.

With the PROFI series, you get a hoist that combines safety, power and precision to elevate your operations. Whether dealing with high demands for cleanliness, explo-

sive atmospheres, or toughest weather conditions the PROFI series remains steadfast, delivering top-tier performance with every lift.

Of course, we ensure you have access to our JDN global service network for your assistance. From technical on-site consultation, to full lifecycle support including JDN original spare parts availability, J.D. Neuhaus guarantees maximum performance, safety and reliability.

Upgrade your equipment portfolio with the JDN PROFI series air hoist. Experience the perfect blend of safety, precision and durability. Elevate your operations and trust in a brand committed to your success.

Choose JDN PROFI. Choose permanent excellence.

INNOVATION AND VALUES

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who work for us, as well as with our partners and customers. They are found in the quality and German engineering of our products, and in the way we work with you and find new solutions thanks to our innovative spirit.

Our thinking and actions are guided by our vision: We are helping people to move loads in a digitized and sustainable world.





APPLICATIONS – LIFTING IN VARIOUS INDUSTRIES

Renowned for its exceptional performance and reliability, the PROFI series excels across a wide range of industries. In shipyards, it plays a crucial role in precisely installing massive and heavy propellers on the world's largest ships, ensuring these giants of the sea are ready for their voyages. In the chemical and pharmaceutical industries, our customers rely on the PROFI series to safely and efficiently lift loads in explosive and humid environments where high cleanliness is required.

On offshore platforms, the PROFI series is indispensable for lifting loads in topside environments, withstanding direct environmental challenges such as harsh weather conditions, saltwater, and explosive atmospheres. In the construction sector, it is highly valued not only for its impressive lifting capacity of up to 100 tons but also for its robustness and mobility, making it an essential tool in tunnelling and bridge building.

In heavy industries such as foundries, the PROFI series is a global favourite, effortlessly handling extreme temperatures and dusty environments with ease. Additionally, in diverse industries like the battery industry, our customers have customized their lifting solutions with unique innovations based on the PROFI series, showcasing the engineering and innovation skills of the JDN team.

No matter the challenge, the PROFI series rises to the occasion, delivering outstanding performance in the most demanding applications. In all the industries it serves, the serviceability and low complexity of its design are among the strongest arguments for the JDN PROFI series.

Experience the PROFI series – engineered for permanent excellence in every industry.

COST EFFICIENT

thanks to easy maintenance

NO DOWNTIME

100% availability

EXPLOSION SAFE

for use in hazardous areas

MAXIMUM SAFETY

in all areas of application

PRECISE

handling of loads

TECHNICAL DATA

PROFI 025 TI TO 2 TI

TYPE	UNIT	025 TI		05 TI		1 TI		2 TI	
Load-carrying capacity	t	0.16	0.25	0.32	0.5	0.63	1	1.25	2
Air pressure	bar	4	6	4	6	4	6	4	6
Number of chain strands		1		1		1		2	
Engine power	kW	0.6	1.0	0.6	1.0	0.6	1.0	0.6	1.0
Lifting speed at rated load	m/min	20	20	10	11	5	5.5	2.5	2.7
Lifting speed without rated load	m/min	30	36	15	17.5	8.5	10	4.5	5
Lowering speed at rated load	m/min	30	38	14	16.5	8	9.7	4	4.8
Air consumption at rated load – lifting	m³/min	1.1	1.7	1.1	1.7	1.1	1.7	1.1	1.7
Air consumption at rated load – lowering	m³/min	1.3	1.9	1.3	1.9	1.3	1.9	1.3	1.9
Air connection		G ½		G ½		G ½		G ½	



PROFI 1,5 TI AND 3 TI/2

TYPE	UNIT	1,5 TI	3 TI/2
Load-carrying capacity	t	1.6	3.2
Air pressure	bar	4 – 6	4 – 6
Number of chain strands		1	2
Engine power	kW	1.3 – 2	1.3 – 2
Lifting speed at rated load	m/min	4.4 – 6.8	2.2 – 3.4
Lifting speed without rated load	m/min	9.1 – 11.0	4.6 – 5.5
Lowering speed at rated load	m/min	12.2 – 13.5	6.0 – 6.7
Air consumption at rated load – lifting	m³/min	1.8 – 3.2	1.8 – 3.2
Air consumption at rated load – lowering	m³/min	2.5 – 4.1	2.5 – 4.1
Air connection		G ¾	G ¾

PROFI 3 TI TO 20 TI

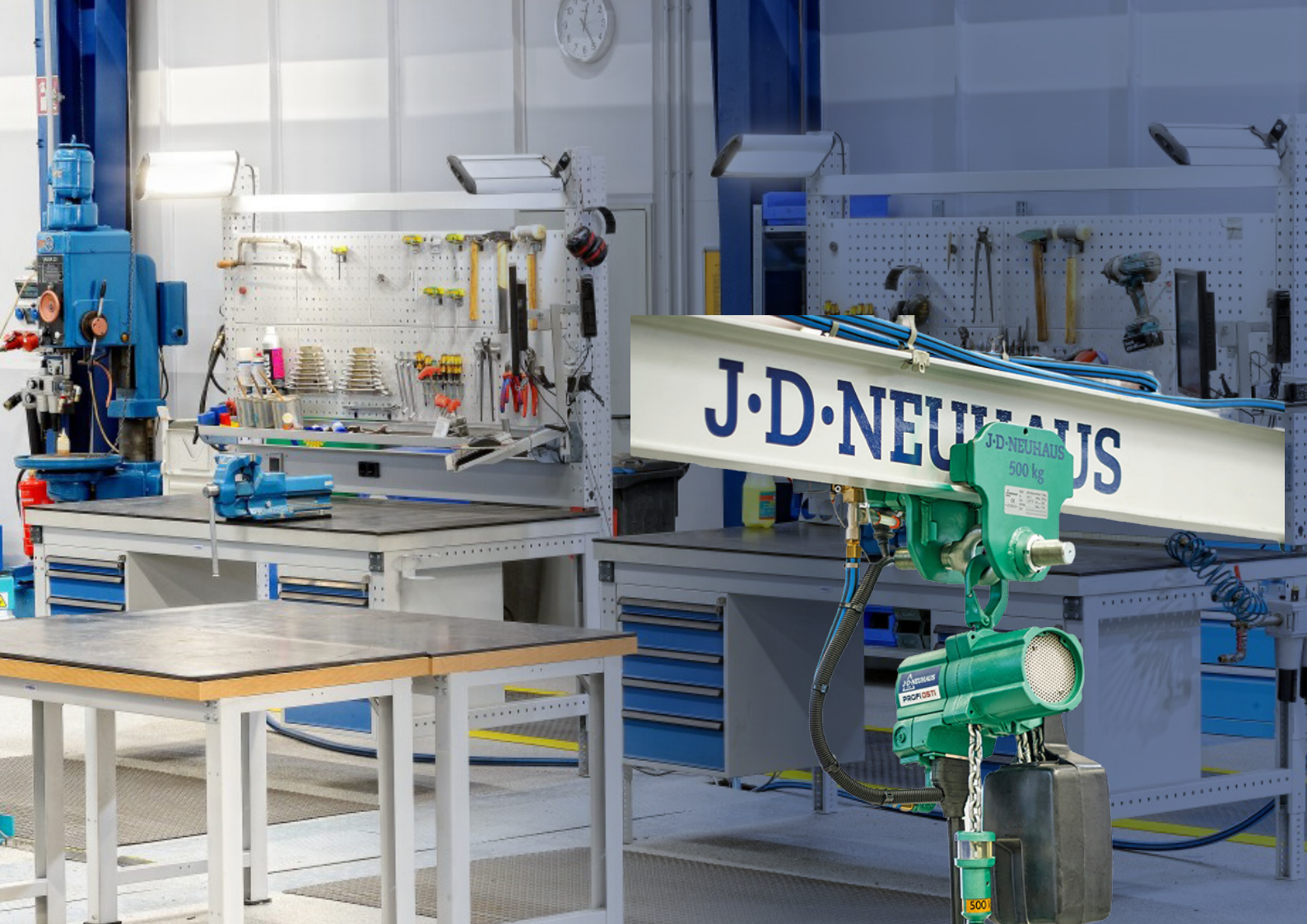
TYPE	UNIT	3 TI		6 TI		10 TI		16 TI		20 TI	
Load-carrying capacity	t	3.2		6.3		10		16		20	
Air pressure	bar	4	6	4	6	4	6	4	6	4	6
Number of chain strands		1		2		2		3		4	
Engine power	kW	1.8	3.5	1.8	3.5	1.8	3.5	1.8	3.5	1.8	3.5
Lifting speed at rated load	m/min	2.5	5.0	1.2	2.5	0.8	1.6	0.5	1.0	0.4	0.7
Lifting speed without rated load	m/min	5.6	9.0	2.8	4.5	1.7	3.0	1.1	2.0	0.7	1.4
Lowering speed at rated load	m/min	5.2	8.4	2.6	4.2	1.5	2.8	1.0	1.9	0.7	1.4
Air consumption at rated load – lifting	m³/min	2.2	4.0	2.2	4.0	2.2	4.0	2.2	4.0	2.2	4.0
Air consumption at rated load – lowering	m³/min	3.0	4.5	3.0	4.5	3.0	4.5	3.0	4.5	3.0	4.5
Air connection		G ¾		G ¾		G ¾		G ¾		G ¾	



PROFI 25 TI TO 100 TI

TYPE	UNIT	25 TI	30 TI	37 TI	40 TI	50 TI	60 TI	75 TI	100 TI
Load-carrying capacity	t	25	30	37	40	50	60	75	100
Air pressure	bar	6							
Number of chain strands		2	2	3	3	4	4	3	4
Engine power	kW	6.3						9	9
Lifting speed at rated load	m/min	1.20	1.00	0.75	0.70	0.55	0.45	0.53	0.40
Lifting speed without rated load	m/min	2.40	2.40	1.70	1.70	1.30	1.30	1.25	0.90
Lowering speed at rated load	m/min	2.80	2.80	1.90	1.90	1.50	1.50	1.25	0.90
Air consumption at rated load – lifting	m³/min	6.5						7.6	
Air consumption at rated load – lowering	m³/min	8.0							
Air connection		G 1 ½							

The stated values refer to standard configurations (without accessories) and room conditions with a flow pressure of 5.8 bar before the hoist. Subject to change without notice.



AIR HOISTS

JDN ENERGY SUPPLIES

A series of supply systems are available for powering JDN Air Hoists in trolleys, monorail hoists and crane systems:

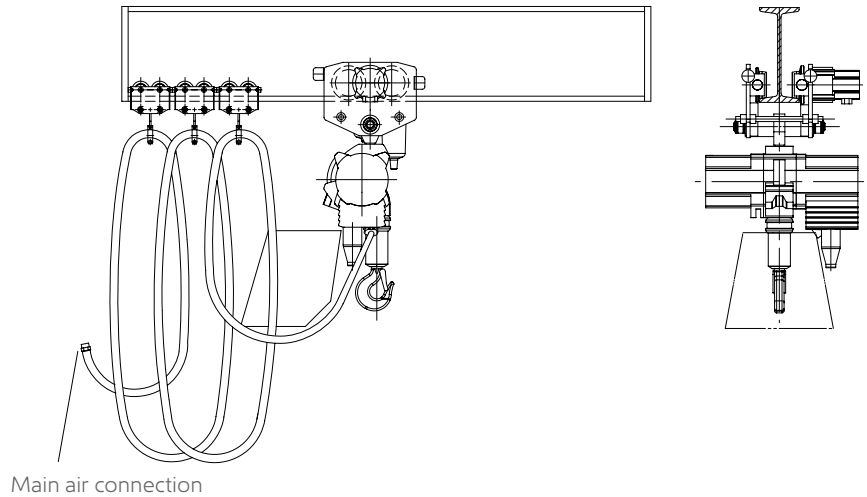
- Hose trolleys
- Spiral hose
- Square bar or C rail
- Energy chain

HOSE TROLLEYS

The hose is fastened to trolleys, which roll directly on the bottom flange of the beam. With each horizontal move of the hoist along the beam, the hose trolleys make the hose follow suit. The hose trolleys will be used for short distances or if there is not enough space on the side of the beam to install C or square bars.

YOUR ADVANTAGES:

- Easy to install
- Cost-efficient
- Consisting of: Hose carriages and hose



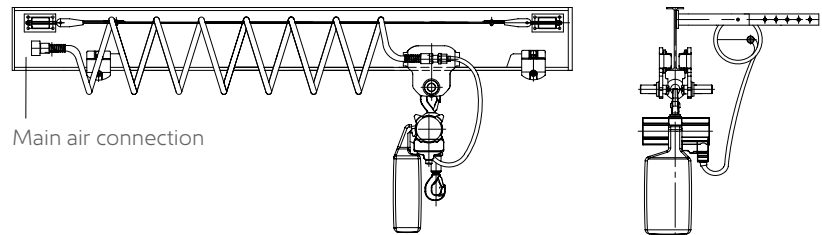
SPIRAL HOSE

This simple and economical solution is suitable for distances of up to 10 metres. The hose rings are suspended on a plastic-coated rope that runs parallel to the track.

The spiral hose can be used in category 3 (zone 2) with gases in explosion group IIA and IIB. It is not suitable for applications in category 2 (zone 1) or group IIC.

Practical tip:

Make sure to lay the hose so that its extended length is roughly 1.5 times the required distance.



- Consisting of: Tensioning arms, rope tensioners, hose and rope

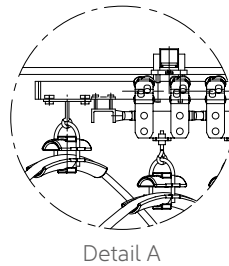
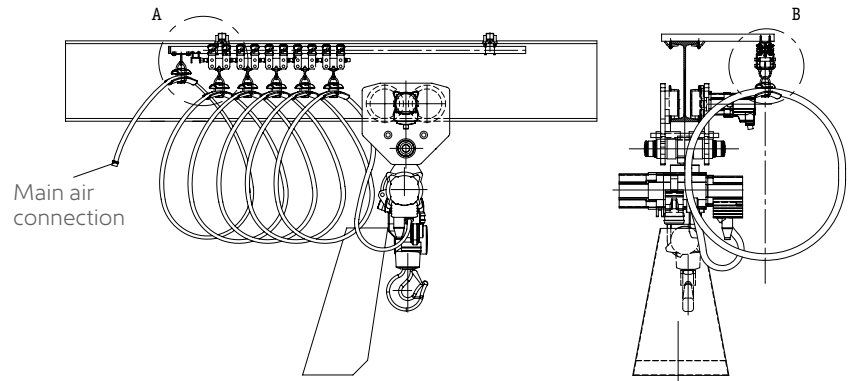
SQUARE BAR & C RAIL

Galvanised C rails or square bars are installed along the beam to carry the energy supply lines.

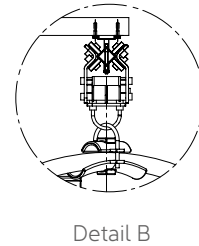
SQUARE BAR

Depending on the local conditions, rails and curved tracks of different lengths are available, as well as an extensive range of installation accessories. The square bar is also suitable for curved tracks.

- Consisting of: Square bars, tensioning arms, hose, hose supports and supply line carriages



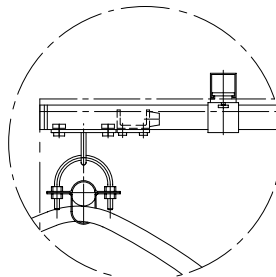
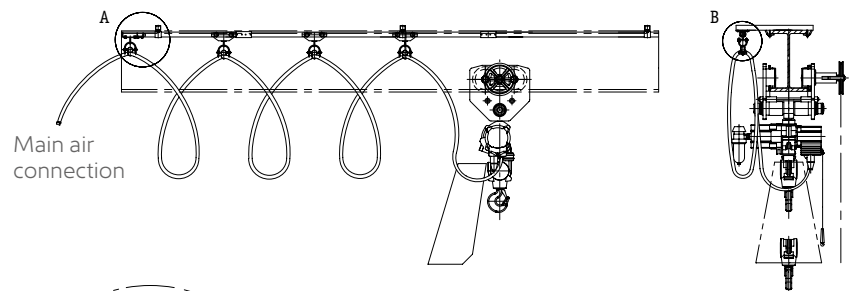
Detail A



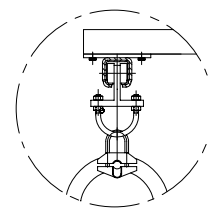
Detail B

C RAIL

- Consisting of: C rails with support, adapter, antistatic supply hose and hose support. The supports must be attached to the upper flange of the beam.



Detail A



Detail B

ENERGY CHAIN

DIE ENERGY CHAIN FOR TROLLEY DRIVE OF OVERHEAD TRAVELLING CRANES

The energy supply for trolley drive in overhead travelling cranes is realised by a horizontal version of the energy chain. A guide channel is mounted on the girder with the chain gliding inside. Air distribution and the control box are also attached to the guide channel. Usually there are two different types of hoses inside the energy chain: The air hose, which feeds the hoisting motor and the trolley motor, and the control hose for crane control functions.

In case of low headroom requirements choose vertical installed energy chain, like the supply for trolleys in underslung cranes.

ENERGY CHAIN FOR TROLLEY AND CRANE DRIVE OF UNDERSLUNG CRANES

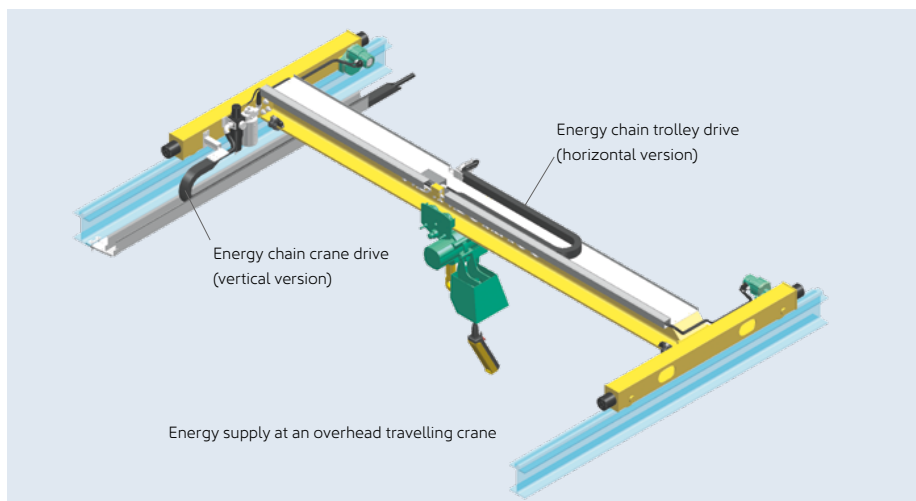
The energy supply for hoists with trolley and for crane drives in underslung cranes is realised by a vertical installation of the energy chain. The guide channel system is mounted with c-consoles, which are clamped by claws on the top flange of the girder. The dimensions of the c-consoles are depending on the used girder.

Supply air hoses for hoisting and travelling as well as control hoses are installed in the energy chain

ENERGY CHAIN FOR CRANE DRIVE OF OVERHEAD TRAVELLING CRANES

The energy supply for crane drive in overhead travelling cranes is realised by a vertical installation of the energy chain. The necessary guide channel system is mounted with clamped brackets on the bottom flange. If different profile sizes for the bottom flange are used because varying in support spacing, the brackets may be clamped to the top flange. The clamped brackets can be used for all the normal steel girder sections (with flange thickness 7-40 mm (0.28 -1.57 inch)).

The energy chain carries air hoses inside, which supply the hoisting motor, driving motors as well as the control hoses of the crane. In addition it is possible to install further control hoses and electric cables inside the energy chain. The main air connection of the energy supply is located midway of the crane travel distance.



CONTACT

For further information, please contact us. Our sales, project management and service staff will be happy to assist you.

J.D. Neuhaus Group

info@jdngroup.com
jdngroup.com

Locations:

Germany - Headquarter, France,
Singapore, United Kingdom, United States





AIR HOISTS

JDN UH ULTRA-LOW MONORAIL HOISTS

CAPACITIES: 1 T UP TO 100 T

CAPACITIES: 1 T UP TO 100 T AIR PRESSURE: 6 BAR

Where loads have to be lifted and transported in extremely reduced spaces the JDN Ultra-Low Monorail Hoists provide the ideal solution. For example the Ultra-Low Monorail Hoist with a load capacity of 6 t has a headroom of only 230 mm.

STANDARD FEATURES

- Ideally suited for working in hazardous areas (explosion risk)
- Insensitive to humidity, dust and temperatures from -20°C up to +70°C
- Extremely low headroom
- Low air consumption
- Available with increased spark protection



INNOVATION AND VALUES

J.D. NEUHAUS – MORE THAN PRODUCTS

As a family-run company in its seventh generation, our commitment to clear values helps us achieve the standard of "Permanent Excellence" that we have held ourselves to for over 275 years.

The three value pairs of tradition and innovation, performance and longevity as well as partnership and passion create the foundation for our daily activities. They shape our partnership with the people who work for us, as well as with our

partners and customers. They are found in the quality and German engineering of our products, and in the way we work with you and find new solutions thanks to our innovative spirit.

Our thinking is guided by sustainability, which we believe our future urgently needs and, of course, deserves.



TECHNICAL DATA

TYPE		UH 1	UH2	UH 4	UH 6	UH 8	UH 12	UH 16	UH 25
Capacity	mt	1	2	4	6	8	12	16	25
	psi	85	85	85	85	85	85	85	85
Air pressure	bar	6	6	6	6	6	6	6	6
Number of chain strands		2 x 1	2 x 2	2	2	4	4	4	4
Motor output	kW	1	1	2.5	2.5	2.5	2.5	2.5	6.3
	ft/min	14.8	7.2	8.2	6.6	4.6	3	2.0	3.6
Lifting speed at full load	m/min	4.5	2.2	3.0	2.0	1.4	0.9	0.6	1.1
	ft/min	30.0	14.8	19.7	14.8	9.5	7.2	3.9	7.9
Lifting speed without load	m/min	9.0	4.5	6.0	4.5	2.9	2.2	1.2	2.4
	ft/min	29.5	14.8	23.0	14.8	10.8	6.9	4.3	7.9
Lowering speed at full load	m/min	9.0	4.5	7.0	4.5	3.3	2.1	1.3	2.4
Air consumption at full load – lifting	cfm	60.0	60.0	141.3	141.3	141.3	141.3	141.3	229.5
	m³/min	1.7	1.7	4.0	4.0	4.0	4.0	4.0	6.5
Air consumption at full load – lowering	cfm	67.1	67.1	158.9	158.9	158.9	158.9	158.9	282.5
	m³/min	1.9	1.9	4.5	4.5	4.5	4.5	4.5	8.0
Air connection		G ½	G ½	G ½	G ½	G ½	G ½	G ½	G ½
Hose dimension (Ø inside)	inch	½	½	¾	¾	¾	¾	¾	1 ½
	mm	13	13	19	19	19	19	19	35
Weight with standard lift	min.		max.						
	lbs	364*	452*	1014	1036	1190	1213	1235	3307
Chain dimension	kg	165*	205*	460	470	540	550	560	1500
	mm	7 x 21	7 x 21	13 x 36	13 x 36	13 x 36	13 x 36	13 x 36	16 x 45
Weight of chain	lbs/ft	0.67	0.67	2.6	2.6	2.6	2.6	2.6	3.9
	kg	1.0	1.0	3.8	3.8	3.8	3.8	3.8	5.8
Standard lift	ft	10	10	10	10	10	10	10	10
	kg	3	3	3	3	3	3	3	3
Length of control at standard lift	ft	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
	m	2	2	2	2	2	2	2	2
Noise level at full load ¹ – lifting	dB(A)	76	76	78	78	78	78	78	80
Noise level at full load ¹ – lowering	dB(A)	78	78	80	80	80	80	80	84

¹Measured at 1 m distance acc. to DIN 45635 part 20
 Group mechanism: M3 (1 Bm)
 Technical data for higher capacities on request.

*Depending on beam width



PNEUMATIC HOISTS

JDN CRANE SYSTEMS

Robust – Reliable – Low maintenance
The individual solution for crane systems
and crane kits

CUSTOM FIT FOR YOUR INDIVIDUAL NEEDS

The delivery programme comprises explosion-proof **Top running overhead travelling cranes, Under hung overhead travelling cranes, Jib cranes**, which can be designed to your individual needs, customised installations are our speciality. Depending on your requirements JDN air hoists in motor trolleys or monorail hoist systems are integrated into the crane design. An ergonomically designed pneumatic pendant control is supplied with two speed control as standard for crane and trolley travel. Infinitely variable hoist and trolley speed control is also available.

We have different JDN cranes in detail:

- Overhead cranes with single or double girder design
- Underhung cranes including low headroom design
- Jib cranes
- Cranes with in line mechanically linked synchronised hoists
- Cranes with parallel operating hoists
- Capacities up to 100 t
- Crane spans up to 36 m
- End carriages with pneumatic drives
- Energy feeding systems
- Safety accessories
- And of course the appropriate air hoist with trolley

Additionally we offer **Crane kits for Explosion-proof air cranes** with capacities up to 10t. J.D. Neuhaus can offer crane manufacturers crane component kits complete with pneumatic crane drives. With these crane kits overhead travelling cranes up to 10 t capacity can be built very simply and economically, especially for applications in hazardous areas. The crane manufacturer provides the main girder and JDN delivers all the components that are necessary to build an air powered crane of their chosen design:

We remain in close contact with our users at all times through our 360° service partnerships – and we use this dialogue to continuously develop our products.

**In short: we keep our promise
“Permanent Excellence” for you.**

INNOVATION AND VALUES

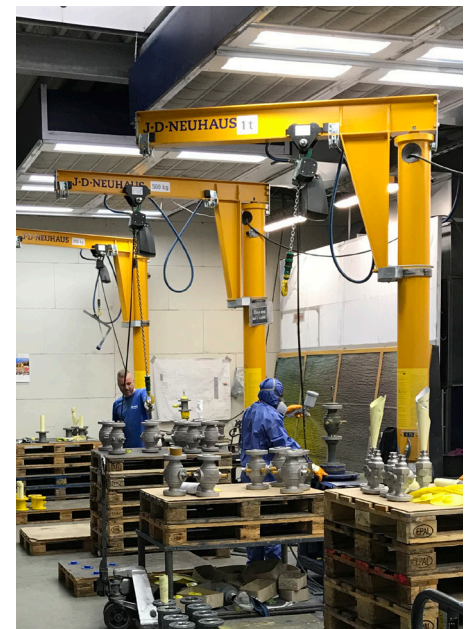
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Our thinking is guided by sustainability, which we believe our future urgently needs and, of course, deserves.





JDN-ACCESSORIES TAILORED TO YOUR INDIVIDUAL NEEDS

We offer a wide range of accessories designed to ensure that JDN standard products are suitable for your specific applications.

This means, for example, that you can meet very specific safety requirements, adjust performance capacity or make operations even more convenient.

- Filter silencer
- Filter regulator
- Service unit
- Main air emergency-stop valve
- Chain box
- Special grease cartridge for oil-free operation, volume 250 ml
- Limit switch for lifting and travelling
- Booster valve (control lengths over 12 m)
- Extension arm for control on motorised trolley
- Additional suspension for chain box (for installation in trolley)
- Copper-plated load hook for increased spark protection
- Stainless steel load hook (up to 750 kg capacity)
- Stainless steel chain (reduced capacity) up to 6 TI
- Manual emergency lowering device for PROFI 3 TI–20 TI hoists
- Special paint finishes

PNEUMATIC

mechanical drive

ROBUST

to external ambient influences

NO DOWNTIME

100% on-time

EXPLOSIONS SAFE

for use in hazardous areas

COST-EFFECTIVE

thanks to low maintenance

Circa 1950s



Circa 1960s



GENERATION UPON GENERATION OF QUALITY

ELEPHANT
LIFTING PRODUCTS™

FAMILY OWNED & OPERATED

Circa 1980s

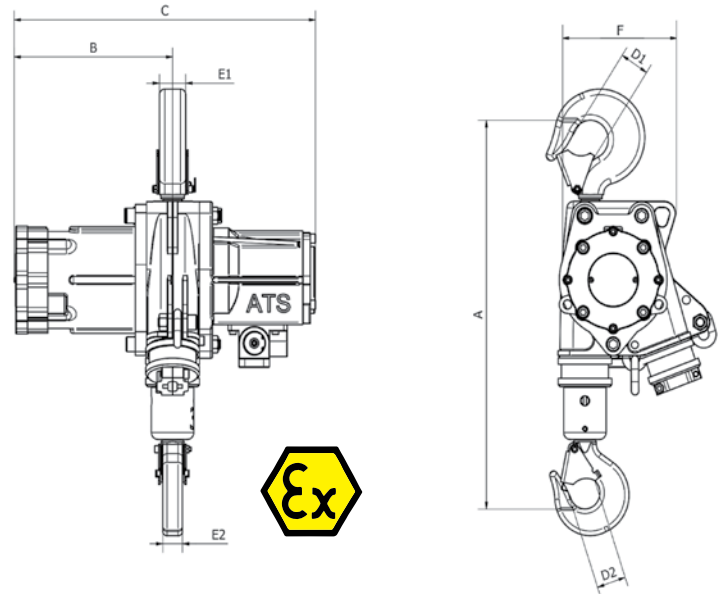


Circa 2000s





ATS INDUSTRIAL SERIES AIR CHAIN HOIST 0.5ton-2ton



SPECIFICATIONS - 0.5ton - 2ton

Lifting capacity	Ton	0.5	1.0	2.0
Model		ATSi-05	ATSi-1	ATSi-2
Number of falls of chain		1	1	1
Weight with 10ft lift	lbs	66	73	93
Chain weight per ft lift	lbs	0.73	0.73	1.2
Chain size	mm	7x21	7x21	9x27
Working air pressure	bar	5-7	5-7	5-7
Air consumption @ full load	CFM	60	75	75
Air hose connection	BSP	3/4"	3/4"	3/4"
Lifting speed @ full load	ft/min	52	16.5	10.5
Lifting speed @ no load	ft/min	98	24	15.2
Lowering speed @ full load	ft/min	82	19	24.3
Noise level	dBA @1m	84	84	84

DIMENSIONS (in) - 0.5ton - 2ton

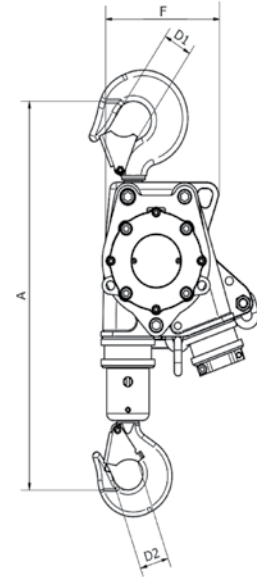
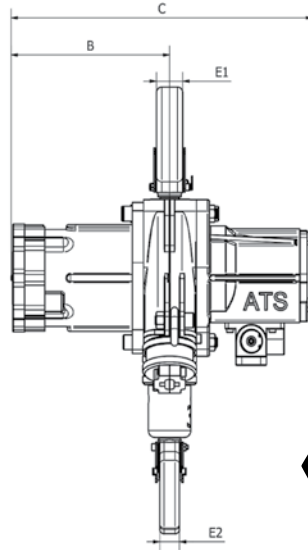
Lifting capacity	0.5ton	1 ton	2ton
A min .headroom	13.8	13.8	14.4
B	3.9	7.0	8.0
C	10.9	13.9	15.1
D1	0.9	0.9	0.9
D2	0.9	0.9	0.9
E1	0.9	0.9	0.9
E2	0.9	0.9	0.9
F	5.6	5.6	5.6

* Chain containers will increase headroom





ATS INDUSTRIAL SERIES AIR CHAIN HOIST 3ton-10ton



SPECIFICATIONS - 3ton - 10ton

Lifting capacity	Ton	3.0	4.2	5.0	6.0	10.0
Model		ATSi-3	ATSi-4	ATSi-5	ATSi-6	ATSi-10
Number of falls of chain		1	1	1	2	2
Weight with 10ft lift	lbs	134	165	179	168	280
Chain weight per ft lift	lbs	1.8	2.6	3.8	3.7	7.7
Chain size	mm	13x36	13x36	16x45	13x36	16x45
Working air pressure	bar	5-7	5-7	5-7	5-7	5-7
Air consumption @ full load	CFM	100	110	110	100	110
Air hose connection	BSP	3/4"	3/4"	3/4"	3/4"	3/4"
Lifting speed @ full load	ft/min	8.9	7.5	5.9	4.4	3.0
Lifting speed @ no load	ft/min	13.1	10.2	9.8	6.6	4.9
Lowering speed @ full load	ft/min	13.8	9.2	11.5	6.9	5.7
Noise level	dBA @1m	84	84	84	84	84

DIMENSIONS (in) - 3ton - 10ton

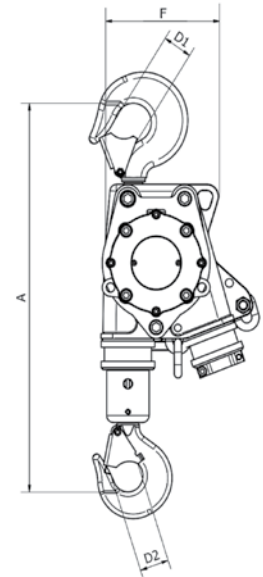
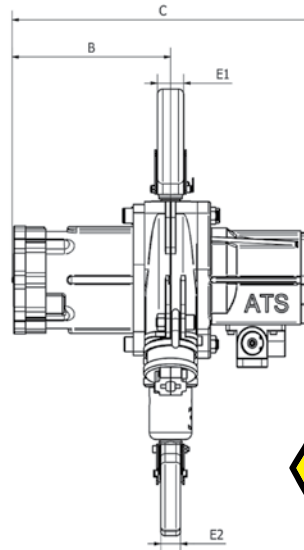
Lifting capacity	3.0	4.2	5.0	6.0	10.0
A min .headroom	21.9	21.9	24.4	27.6	31.3
B	9.4	10.7	10.8	9.4	10.8
C	18.0	19.3	19.6	18.0	19.6
D1	1.2	1.2	1.6	1.6	1.7
D2	1.6	1.6	1.6	1.6	1.7
E1	1.8	1.8	1.9	1.9	2.1
E2	1.9	1.9	1.9	1.9	2.1
F	9.33	9.33	9.33	9.33	10.04

* Chain containers will increase headroom





ATS INDUSTRIAL SERIES AIR CHAIN HOIST 16ton-25ton



SPECIFICATIONS - 16ton - 25ton

Lifting capacity	Ton	16.0	18.0	20.0	25.0
Model		ATSi-16	ATSi-18	ATSi-20	ATSi-25
Number of falls of chain		4	3	4	4
Weight with 10ft lift	lbs	392	728	874	941
Chain weight per ft lift	lbs	10.2	11.5	15.3	15.3
Chain size	mm	13x36	16x45	16x45	16x45
Working air pressure	bar	5.7	5-7	5-7	5-7
Air consumption @ full load	CFM	110	110	110	110
Air hose connection	BSP	3/4"	3/4"	3/4"	3/4"
Lifting speed @ full load	ft/min	2.0	3.0	2.1	1.8
Lifting speed @ no load	ft/min	2.6	4.3	4.6	4.6
Lowering speed @ full load	ft/min	2.3	4.3	5.9	5.9
Noise level	dBA @1m	84	84	84	84

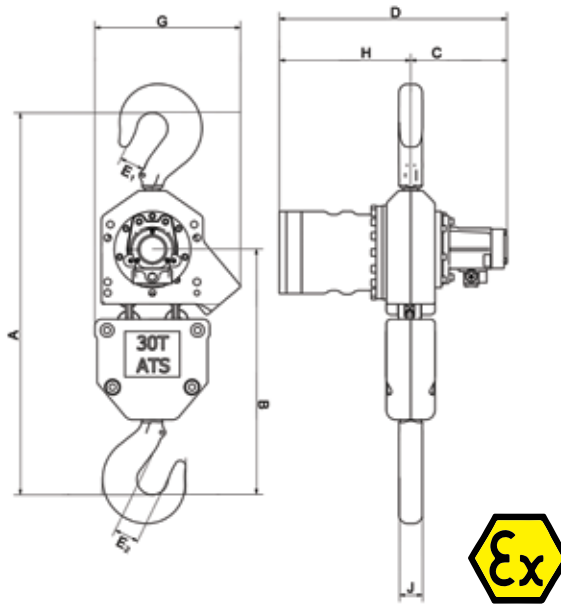
DIMENSIONS (in) - 16ton - 25ton

Lifting capacity	16.0	18.0	20.0	25.0
A min .headroom	31.3	39.8	39.8	42.5
B	12.6	13.5	12.9	12.9
C	23.7	23.9	24.3	24.3
D1	2.3	3.1	3.1	2.8
D2	2.3	3.1	3.1	2.8
E1	2.0	2.5	2.5	3.0
E2	2.0	2.5	2.5	3.0
F	12.4	12.4	13.0	13.0

* Chain containers will increase headroom



ATS INDUSTRIAL SERIES AIR CHAIN HOIST 30ton-50ton


SPECIFICATIONS - 30ton - 50ton

Lifting capacity	Ton	30.0	45.0	50.0
Model		ATSi-30	ATSi-45	ATSi-50
Number of falls of chain		2	3	4
Weight with 10ft lift	lbs	1,120	1,960	2,688
Chain weight per ft lift	lbs	16.4	24.6	32.8
Chain size	mm	23,5x66	23,5x66	23,5x66
Working air pressure	bar	5-7	5-7	5-7
Air consumption @ full load	CFM	230	230	225
Air hose connection	BSP	1 1/2"	1 1/2"	1 1/2"
Lifting speed @ full load	ft/min	3.9	2.6	1.6
Lifting speed @ no load	ft/min	8.5	4.9	4.3
Lowering speed @ full load	ft/min	6.6	4.9	3.3
Noise level	dBA @1m	84	84	84

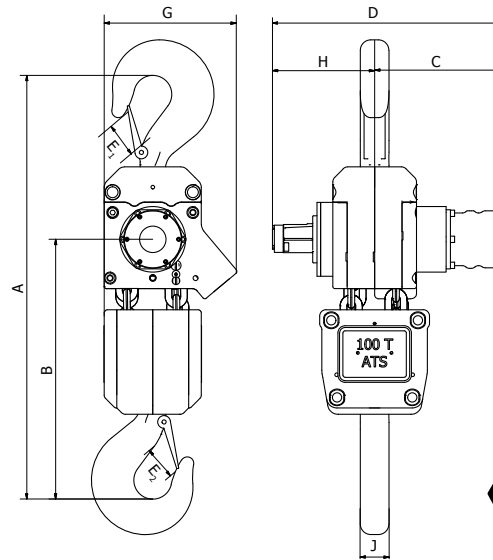
DIMENSIONS (in) - 30ton - 50ton

Lifting capacity	30.0	45.0	50.0
A min .headroom	51.6	59.1	59.1
B	14.1	18.1	17.3
C	28.5	34.8	34.8
D1	2.8	3.9	3.9
D2	2.8	3.9	3.9
E1	3.0	4.2	4.2
E2	3.0	4.2	4.2

* Chain containers will increase headroom



ATS INDUSTRIAL SERIES AIR CHAIN HOIST 60ton-100ton



SPECIFICATIONS - 60ton - 100ton

Lifting capacity	Ton	60.0	75.0	100.0
Model		ATSi-60	ATSi-75	ATSi-100
Number of falls of chain		4	3	4
Weight with 10ft lift	lbs	2,688	4,256	5,376
Chain weight per ft lift	lbs	32.8	45.6	60.7
Chain size	mm	23,5x66	32x90	32x90
Working air pressure	bar	5-7	5-7	5-7
Air consumption @ full load	CFM	230	280	280
Air hose connection	BSP	1 1/2"	1 1/2"	1 1/2"
Lifting speed @ full load	ft/min	1.5	2.2	1.6
Lifting speed @ no load	ft/min	4.3	3.1	2.3
Lowering speed @ full load	ft/min	3.3	3.5	2.6
Noise level	dBA @1m	84	84	84

DIMENSIONS (in) - 60ton - 100ton

Lifting capacity	60.0	75.0	100.0
A min .headroom	59.1	70.1	77.2
B	17.3	25.2	23.8
C	34.8	43.5	43.5
D1	3.9	4.9	4.9
D2	3.9	4.9	4.9
E1	4.2	5.2	5.2
E2	4.2	4.2	5.2

* Chain containers will increase headroom





AIR CHAIN HOIST WITH MOTORIZED TROLLEY

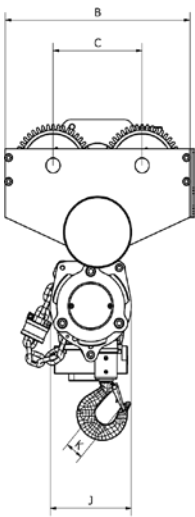
- Motorized trolleys available for the ATSI & Rigger Series
- For hoist capacities greater than 20 ton, see the Monorail Style Units
- Air motor has a polished stainless steel cylinder
- Anti-tip and anti-drop devices
- Manufactured with high quality steel components, allowing for a compact design
- Available with spark resistant features
- Available with marine paint finish

SPECIFICATIONS - ATS MOTORIZED TROLLEYS - 0.5 ton - 5ton

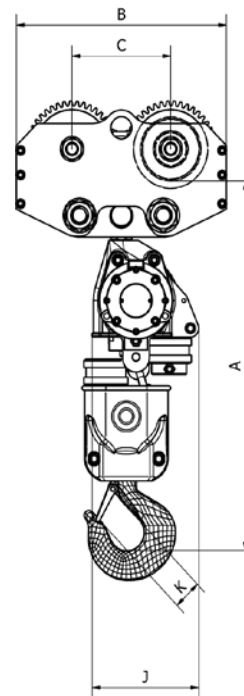
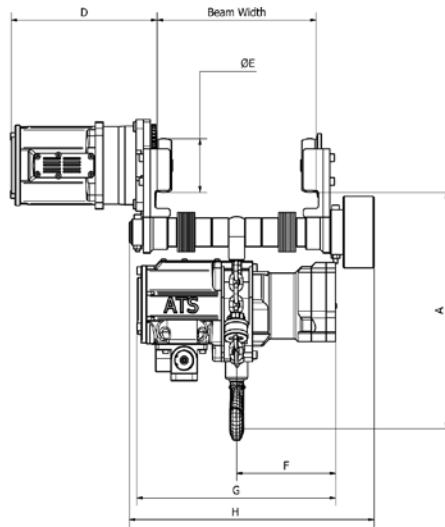
Lifting capacity	Ton	0.5	1	2	3	4.2	5
Model		ATSM-0.5	ATSM-1	ATSM-2	ATSM-3	ATSM-4.2	ATSM-5
Number of falls of chain		1	1	1	1	1	1
Weight with 10ft lift	lbs	67.2	73.9	94.1	134.4	168.0	179.2
Chain weight per ft lift	lbs	0.73	0.73	1.18	1.8	2.6	3.8
Chain size	mm	7x21	7x21	9x27	13x36	13x36	16x45
Working air pressure	bar	5-7	5-7	5-7	5-7	5-7	5-7
Air consumption @ full load	CFM	60	75	75	100	110	110
Air hose connection	BSP	3/4"	3/4"	3/4"	3/4"	3/4"	3/4"
Lifting speed @ full load	ft/min	52	16.5	15	8.9	7.5	5.9
Lifting speed @ no load	ft/min	98	24	28	13.1	10.2	9.8
Lowering speed @ full load	ft/min	82	19	30	13.8	9.2	11.5
Trolley Travelling Speed	ft/min	39.4	39.4	39.4	39.4	32.8	32.8
Noise level	dBA @3ft	84	84	84	84	84	84

* Increased speed conversion kits are available

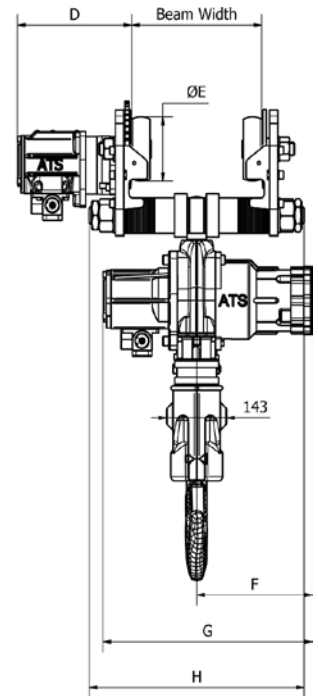




*Schematics for reference only



*Schematics for reference only



DIMENSIONS (in.) - ATS MOTORIZED TROLEYS - 0.5ton - 5ton						
Lifting Capacity (t)	0.5	1.0	2.0	3.0	4.2	5.0
Beam width range	3.5-8.3				4.3-8.3	
A min. headroom LUG	15.6	15.6	15.8	22.8	24.2	24.4
A min. headroom HOOK	19.6	19.6	20.2	29.9	33.0	33.0
B	12.8	12.8	12.8	13.2	14.6	14.6
C	6.2	6.2	6.2	6.3	7.0	7.0
D	10.1	10.1	10.1	10.3	10.8	10.8
E	3.5	3.5	3.5	3.6	4.4	4.4
F	4.0	6.9	8.0	9.6	10.9	11.1
G	10.9	13.8	15.1	18.3	19.6	20.0
H	13.8	13.8	13.8	13.0	14.8	14.8
J	5.6	5.6	5.6	9.3	9.3	12.2
K	0.9	0.9	0.9	1.2	1.2	1.6





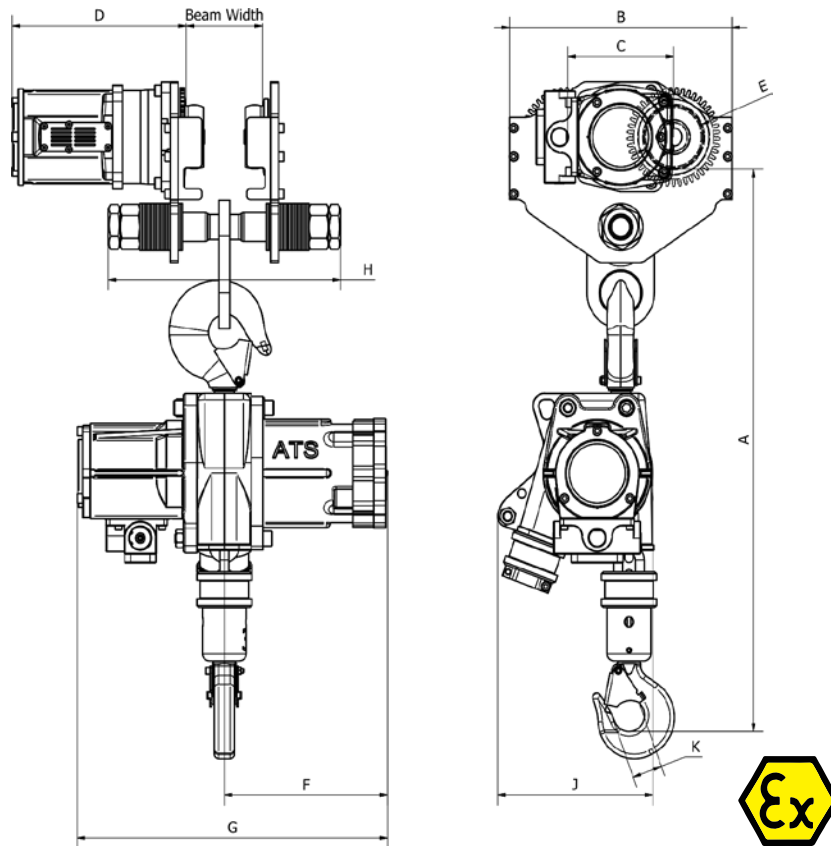
AIR CHAIN HOIST WITH MOTORIZED TROLLEY

SPECIFICATIONS - ATS MOTORIZED TROLLEYS - 6ton - 20ton

Lifting capacity	Ton	6	10	16	18	20
Model		ATSM-6	ATSM-10	ATSM-16	ATSM-18	ATSM-20
Number of falls of chain		2	2	4	3	4
Weight with 10ft lift	lbs	168.0	280.0	1299.1	728.0	873.6
Chain weight per ft lift	lbs	3.7	7.7	10.2	11.5	15.3
Chain size	mm	13x36	16x45	13x36	16x45	16x45
Working air pressure	bar	5-7	5-7	5-7	5-7	5-7
Air consumption @ full load	CFM	100	110	110	110	110
Air hose connection	BSP	3/4"	3/4"	3/4"	3/4"	3/4"
Lifting speed @ full load	ft/min	4.4	3.0	2.0	3.0	2.1
Lifting speed @ no load	ft/min	6.6	4.9	2.6	4.3	4.6
Lowering speed @ full load	ft/min	6.9	5.7	2.3	4.3	5.9
Trolley Travelling Speed	ft/min	32.8	26.2	26.2	26.2	26.2
Noise level	dBA @3ft	84	84	84	84	84

* Increased speed conversion kits are available





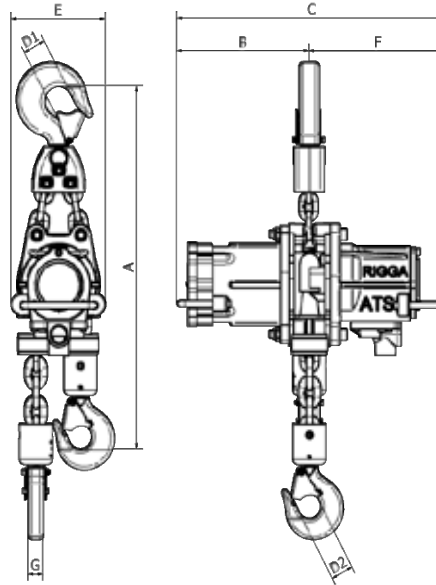
*Schematics for reference only

DIMENSIONS (in.) - ATS MOTORIZED TROLEYS - 0.5ton - 5ton					
Lifting Capacity (t)	6.0	10.0	16.0	18.0	20.0
Beam width range	4.3-8.3	6.7-12.2	6.9-13.0		
A min. headroom LUG	29.3	30.4	34.3	41.9	41.9
A min. headroom HOOK	36.1	38.2	42.9	50.0	50.0
B	14.6	20.0	23.6	24.0	24.0
C	7.0	9.4	9.3	11.4	11.4
D	10.8	10.9	10.9	11.3	11.3
E	4.4	5.8	5.7	7.9	7.9
F	9.6	11.1	12.6	13.6	12.9
G	18.3	20.0	23.7	23.9	24.3
H	14.8	19.6	23.2	21.7	21.7
J	9.3	12.2	11.8	13.0	13.0
K	1.6	2.1	2.3	3.1	3.1





RIGGER SERIES AIR CHAIN HOIST



SPECIFICATIONS - ATS RIGGER CHAIN HOIST - 1ton - 8ton

Lifting capacity	t	1/1	2/2	3/3	3/6	4,2/4,2	5/5	4,2/8,4
Number of falls of chain		1/1	1/1	1/1	1/2	1/1	1/1	1/2
Weight with R-Control	lbs.	81.6	81.6	125.7	178.6	156.5	183.0	187.4
Weight with D-Control	lbs.	92.6	103.6	131.2	185.2	163.1	176.4	194.0
Chain size	mm	9x27	9x27	11,2x34	11,2x35	13x36	16x45	13x36
Chain weight per m lift	lbs./ft	1.2	1.2	1.8	1.8	2.6	3.9	2.6
Working air pressure	bar	4						
Air consumption @ full load	CFM	61.8	61.8	61.8	61.8	61.8	61.8	61.8
Air hose connection	BSP	1"						
Lifting speed @ full load	ft/min	16.4	9.8	14.8	14.8/7.4	10.5	8.5	5.2
Lifting speed @ no load	ft/min	55.8	32.8	19.0	16.4/8.2	13.1	10.5	143904.2
Lowering speed @ full load	ft/min	65.6	42.7	20.3	24.6/12.5	14.8	8.5	7.4
Standard lift	ft	10						
Length of control for Standard Lift	ft	6.5						
Noise level	dBA@3ft	84						

DIM. (in.) - ATS RIGGER - 1ton - 8ton

Type	1 ton	2 ton	3 ton	4,2 ton	5 ton	6 ton	8 ton
A min .headroom	12.8	12.8	24.0	20.6	28.7	2.6	28.7
B	4.7	4.7	8.9	8.9	12.6	13.1	13.1
C	8.0	8.0	9.2	9.2	10.5	9.2	10.5
D	15.0	15.0	18.9	18.9	20.5	18.9	20.5
E1	1.3	1.3	1.6	1.6	1.7	1.6	1.7
E2	1.3	1.3	1.6	1.6	1.7	1.6	1.7
G max.width	5.7	5.7	9.4	9.4	13.4	9.4	13.4
H	7.1	7.1	9.7	9.7	10.0	9.7	10.0
I	8.3	8.3	11.8	11.8	18.9	16.7	18.9
J Hook width	0.9	0.9	1.8	1.8	2.1	1.9	2.1

* Chain containers will increase headroom





ROPE CONTROL

Pull cord control allows for infinitely variable speed for lifting and lowering applications.



PENDANT CONTROL

Pendant controls are machined from solid brass for corrosion resistance. The control's emergency stop button is a standard feature. This style pendant control is also available in aluminum.



REMOTE CONTROL PANEL

With lift synchronization that allows for operation of any one hoist, or a combination of multiple units simultaneously



WIRELESS CONTROL PENDANT

Offered for standard environments, as well as spark resistant applications.



Standard Model ATS Air Chain Hoists are Acceptable for Division 1 & 2 Applications, as follows:

- Division 1 (Spark-Resistant air hoists are required)
Atmospheres containing hazardous concentrations of flammable gases, dusts, or fibers, continuously, frequently or periodically, under normal operating conditions.
- Division 2 (Standard air hoists are required)
Atmospheres, normally non-hazardous, in which a hazardous concentration of flammable gases, dusts, or fibers, occurs when a container or handling system fails, or the ventilating system which normally keeps the concentration non-hazardous, fails

Classification of hazardous places

Gases, Vapours

Zone 0

place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is present continuously or for long periods or frequently

Zone 1

place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is likely to occur in normal operation occasionally

Zone 2

place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is not likely to occur in normal operation but, if it does occur, will persist for a short period only

Dusts

Zone 20

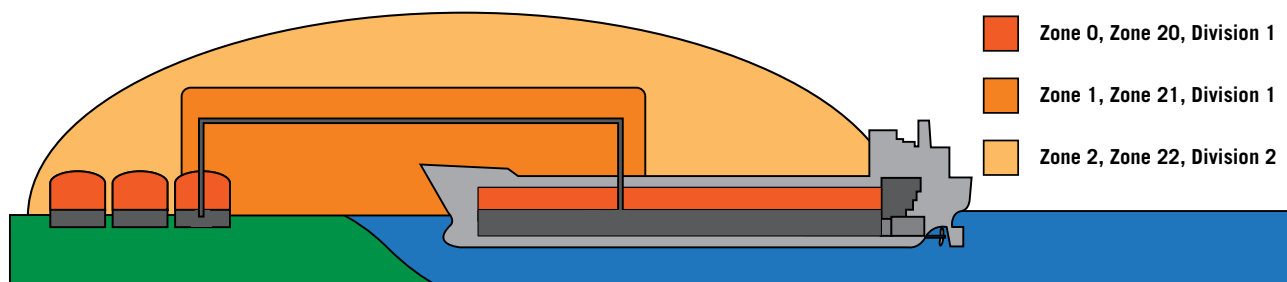
place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently

Zone 21

area in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur, occasionally in normal operation

Zone 22

area in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only

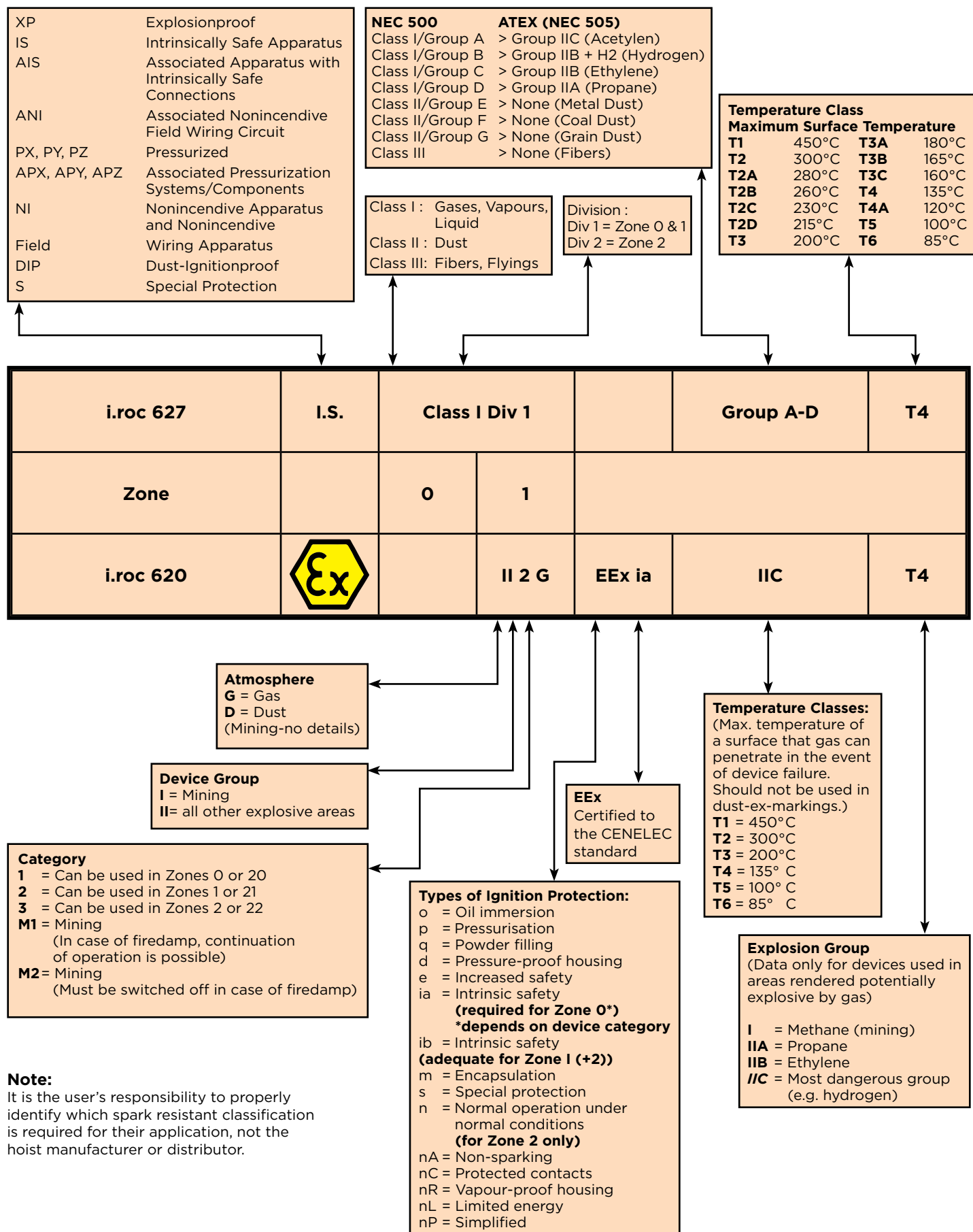


The above rendition is for reference only

Hazardous area				
Conditions and subdivisions			Required marking on the usable equipment	
Flammable materials	Temporary behaviour of explosive atmosphere	Classification of hazardous areas	Equipment Group as defined in directive 94/9/EC	Equipment category as defined in directive 94/9/EC
gases vapours	is present continuously or for long periods or frequently	zone 0	II	1G
	arises in normal operation occasionally	zone 1	II	2G or 1G
	is not likely to arise in normal operation, or if it does, will persist for a short time only	zone 2	II	3G or 2G or 1G
dusts	is present in the form of a cloud continuously, or for long periods or frequently	zone 20	II	1D
	occasionally develops into a cloud during normal operation	zone 21	II	2D or 1D
	is not likely to develop into a cloud during normal operation, or if it does, for a short time only	zone 22	II	3D or 2D or 1D
methane carbon dust	eration where there is a risk of explosion	-	I	M1
	disconnection where there is a risk of explosion	-	I	M2 or M1

Above codes are applicable to Atex standards

ATEX Flow Chart



Note:

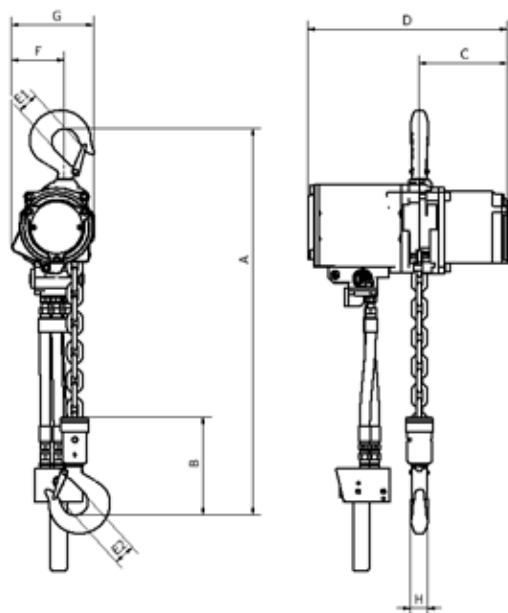
It is the user's responsibility to properly identify which spark resistant classification is required for their application, not the hoist manufacturer or distributor.



ATS COMPACT SERIES AIR CHAIN HOIST 0.25t ~ 2 ton

SPECIFICATIONS - ATSC COMPACT SERIES AIR CHAIN HOIST - 0.5t - 2 ton				
Lifting capacity	Ton	0.5	1	2
Model		ATSi-05C	ATSi-1C	ATSi-2C
Number of falls of chain		1	1	2
Weight with 10ft lift	lbs.	38.1	43.7	53.8
Chain weight per ft lift	lbs.	2.4	2.4	4.8
Chain size	mm	7x21	7x21	7x21
Working air pressure	bar	5-7	5-7	5-7
Air consumption @ full load	CFM	60	60	60
Air hose connection	BSP	1/2"	1/2"	1/2"
Lifting speed @ full load	ft/min	32.8	21.7	13.1
Lifting speed @ no load	ft/min	55.8	32.8	19.0
Lowering speed @ full load	ft/min	65.6	42.7	20.3
Noise level	dBA @3ft	84	84	84





Features:

- Rust free & replaceable stainless steel cylinder
- 100% Duty cycle
- Robust epicycle gearbox (grease filled and sealed)
- Choice pilot pendant with E-Stop or direct pendant control - Variable speed
- Urethane yielding buffers for upper and lower limit stops
- Aluminium and SG Iron housings
- Internal silencing
- High grade carbon steel swivel hooks with safety catches
- Galvanized Grade 80 Load chain made to EN818-7 standard – 5:1 Design factor of safety
- Automatic self-adjusting multi disc brake 125% WLL
- Tried and tested delta P design load limiting device set 130% WLL*
- Spark resistant (Atex Zone 2)

Options:

- Marine spec - Nickel coated and marine paint
- Corrosion resistance: stainless steel hooks and chain, special paints
- Chain containers
- Plain, geared and motorized trolleys

DIMENSIONS (in.) - ATSC COMPACT SERIES AIR CHAIN HOIST - 0.25t - 2 ton

Type	0.5t	1t	2t
A min. headroom	12.1	12.1	15.6
B	5.6	5.6	7.9
C	3.5	5.0	8.0
D	9.9	11.4	15.0
E1	0.9	0.9	0.9
E2	0.9	0.9	0.9
G max.width	4.7	4.7	5.6
H Hook Width	0.9	0.9	0.9



ATS MONORAIL HOISTS

10ton - 60ton



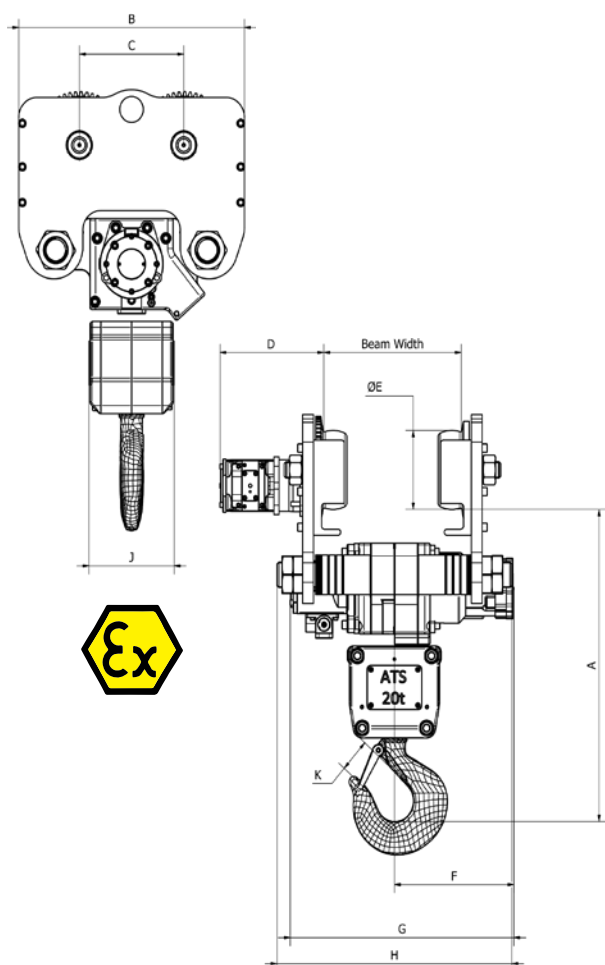
Standard Features:

- Standard Hoists are Atex approved, EX II 2 GD IIC T4 (X) / EX II 2 GD IIB T4 (X) / EX II 2 GD IIC T4 (X)
- Copper platings and stainless steel components can be offered for increased spark resistance
- Rust free & replaceable Stainless steel cylinder
- 100% Duty cycle
- Ideal for working in hazardous areas (Explosive environments)
- Temperature Rated -4°F to 158°F
- Insensitive to dust and humidity
- High Grade carbon steel swivel hooks with safety latches
- Galvanized Grade 80 Load chain made to EN818 standard - 5:1 Design factor of safety
- Automatic self-adjusting multi disc brake 125% WLL.
- Tried and tested slip clutch design load limiting device set 130% WLL.
- Compact design and Lowest headroom in class.
- Anti-climb and Anti-drop devices standard.
- Pilot Pendant control with Emergency shut-off valve
- Complaint with ASME B30.16

SPECIFICATIONS - ATS MONORAIL HOISTS - 10ton - 100ton

		t	10.0	16.0	25.0	30.0	45.0	50.0	60.0	75.0	100.0
Lifting capacity	t		10.0	16.0	25.0	30.0	45.0	50.0	60.0	75.0	100.0
Number of falls of chain			2	3	4	2	3	4	4	3	4
Weight with 3m lift	lbs.		573	937	1,510	1,863	3,219	3,638	3,638	7,937	12,125
Chain weight per m lift	lbs./ft		7.7	11.5	15.3	16.4	24.6	32.8	32.8	45.6	60.7
Chain size	mm		16x45	16x45	16x45	23,5x66	23,5x66	23,5x66	23,5x66	32x90	32x90
Working air pressure	bar		5-7	5-7	5-7	5-7	5-7	5-7	5-7	5-7	5-7
Air consumption @ full load	CFM		110	110	110	230	230	230	230	230	230
Air hose connection	BSP		3/4"	3/4"	3/4"	1 1/2"	1 1/2"	1 1/2"	1 1/2"	1 1/2"	1 1/2"
Lifting speed @ full load	ft/min		4.3	3.0	1.8	3.9	2.6	2.1	2.0	2.0	1.5
Lifting speed @ no load	ft/min		9.2	6.2	4.6	8.5	5.6	4.3	4.3	3.1	2.3
Lowering speed @ full load	ft/min		11.8	7.9	6.1	7.2	8.1	3.3	3.6	3.9	3.0
Traveling speed	ft/min		18.0	18.0	18.0	18.0	18.0	18.0	18.0	18.0	18.0
Rack drive travelling speed	ft/min		23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0	23.0
Standard lift	ft		10	10	10	10	10	10	10	10	10
Length of control for standard lift	ft		6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
Noise level @3ft	dBA @3ft		84	84	84	84	84	84	84	84	84





ATS Monorail hoists are designed for the offshore industry, or wherever heavy loads have to be moved in confined spaces. Depending on the application these hoists can be used in pairs or even sets of four units. For example: Working in parallel units handling BOP handling systems from 20 ton to 150 tons. Or working in unison and connected by a tie bar for handling heat exchangers.

Options:

- Rack and pinion drive
- Marine Spec - Nickel coated and marine paint
- Corrosion resistance: special paints
- Chain containers
- Integrated Wireless Load cells
- Remote control panel - with lift synchronization
- Third Party Acceptance available on request

Special Application:

- If you have a unique application and require a non-standard system, a unit can be designed to your requirements

DIMENSIONS (in.) - ATS MONORAIL HOISTS - 10ton - 100ton

Lifting Capacity (t)	10.0	16.0	25.0	30.0	45.0	50.0	60.0	75.0	100.0
Beam width range	6.3-15.6	5.9-12.2	6.7-13.0	6.9-15.7	5.9-15.2	5.5-14.6	5.5-14.6	6.3-13.8	
A min .headroom	27.6	27.6	32.0	37.2	43.3	46.5	46.5	44.5	59.4
B	19.8	22.1	24.6	24.6	56.3	56.3	56.3	74.8	107.3
C	9.3	9.3	11.4	11.4	11.8	11.8	11.8	13.6	13.6
D	10.7	10.7	11.3	11.3	11.3	11.3	11.3	12.1	12.1
E	5.7	5.7	7.9	7.9	7.9	7.9	7.9	9.6	9.6
F	10.9	13.0	11.5	14.3	17.3	17.3	17.3	17.4	24.6
G	19.7	23.9	23.3	28.5	34.8	34.8	34.8	34.8	45.7
H	19.3	21.3	22.8	21.7	28.5	28.5	28.5	28.5	
J	12.0	7.2	9.3	7.0	11.6	14.8	14.8	19.4	18.1
K	2.1	2.3	2.8	2.8	2.8	3.9	3.9	4.9	4.9





ATS LOW PROFILE 0.25ton - 6ton

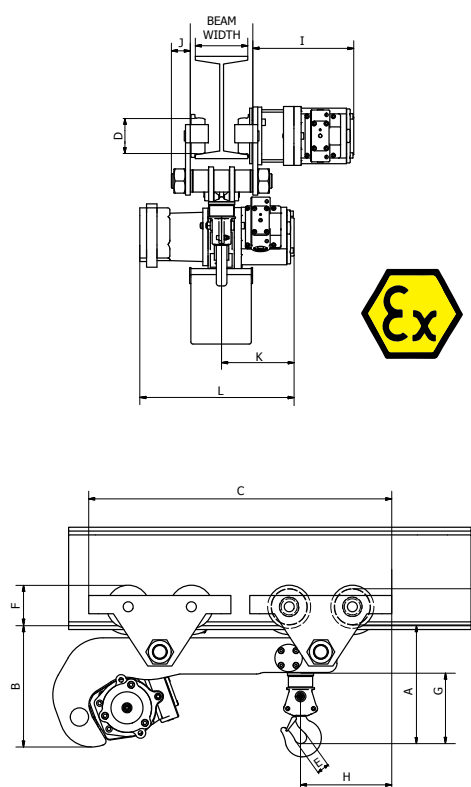
Ideal for areas where headroom is restricted and standard trolleys are not able meet the lifting height requirements, the ATS UL Series Low Headroom Trolleys are recommended.

SPECIFICATIONS - ATS ULTRA LOW PROFILE 0.25ton - 6ton

		ATSUL-025	ATSUL-050	ATSUL-1	ATSUL-2	ATSUL-3	ATSUL-4.2	ATSUL-5	ATSUL-6
Trolley Model / Trolley Carrying Capacity		LM-2, 2ton				LM-6, ton			
Carrying Capacity of Hoist with Trolley	mt	0.25	0.5	1	2	3	4.2	5	6
	T	0.27	0.55	1.1	2.2	3.3	4.6	5.5	6.6
Unit Weight with 10 foot lift	lbs.	198	198	211	219	465	496	498	510
Weight of Chain per additional foot of lift	lbs. / ft.	2.2	2.2	2.2	4	6	8.3	13	12
Chain Size		7x21			9x27	13x36	13x36	16x45	13x36
Number of chain strands		1							2
Air Pressure Trolley Motor	bar	4 ~ 6							
Air Consumption	cfm	60-75			95	110			
Motor Power Output (Hoist)	kW	1.4				3.2			
	H.P.	1.8				4.2			
Lifting Speed at Full Load	ft./min	88.5	64.0	22.0	29.0	15.0	12.0	8.5	7.5
Lifting Speed without Load		121.0	121.0	36.0	43.0	19.0	16.0	18.0	10.0
Lowering Speed with Full Load		82.0	82.0	29.5	29.5	20.0	19.0	16.0	10.0
Traveling Speed	ft./min	19				39			
Air Hose Connection		3/4"							
Air Hose Dimension		3/4"							
Traversing I-Beam Width	inch	3.3 ~ 12.2				1.3 ~ 4.9			
Maximum Bottom Flange Thickness	inch	0.98				1.3			
Noise Level at Motorized Trolley	dB(A)	84							

* Increased speed conversion kits are available





Standard Features:

- Standard Hoists are Atex approved, EX II 2 GD IIC T4 (X) / EX II 2 GD IIB T4 (X) / EX II 2 GD IIC T4 (X)
- Copper platings and stainless steel components can be offered for increased spark resistance
- Rust free & replaceable Stainless steel cylinder
- 100% Duty cycle
- Ideal for working in hazardous areas (Explosive environments)
- Temperature Rated -4°F to 158°F
- Insensitive to dust and humidity
- High Grade carbon steel swivel hooks with safety catches.
- Galvanized Grade 80 Load chain made to EN818 standard - 5:1 Design factor of safety
- Automatic self-adjusting multi disc brake 125% WLL.
- Tried and tested slip clutch design load limiting device set 130% WLL.
- Compact design and Lowest headroom in class.
- Anti-climb and Anti-drop devices standard.
- Pilot Pendant control with Emergency shut-off valve
- Compliant with ASME B30-11 and B30-16

Options:

- Marine Spec - Nickel coated and marine paint
- Corrosion resistance: special paints
- Chain containers

DIMENSIONS - ATS ULTRA LOW PROFILE 0.25ton - 6ton

		ATSUL-050	ATSUL-1	ATSUL-2	ATSUL-3	ATSUL-4.2	ATSUL-5	ATSUL-6
A	mm	267	267	270	345	376	433	522
	inch	10.5	10.5	10.6	13.6	14.8	17.0	20.6
B	mm	305	305	301	280	280	280	280
	inch	12.0	12.0	11.85	11.0	11.0	11.0	11.0
C	mm	680	680	680	845	845	845	845
	inch	26.8	26.8	26.8	33.3	33.3	33.3	33.3
D	mm	90	90	90	109	109	109	109
	inch	3.5	3.5	3.5	4.3	4.3	4.3	4.3
E	mm	23	23	23	30	30	40	40
	inch	0.9	0.9	0.9	1.2	1.2	1.6	1.6
F	mm	122	122	122	154	154	154	154
	inch	4.8	4.8	4.8	6.1	6.1	6.1	6.1
G	mm	145	145	148	230	230	268	357
	inch	5.7	5.7	5.8	9.1	9.1	10.6	14.1
H	mm	207	207	207	168	168	168	168
	inch	8.1	8.1	8.1	6.6	6.6	6.6	6.6
I	mm	254	254	254	272	272	272	272
	inch	10.0	10.0	10.0	10.7	10.7	10.7	10.7
J	mm	58	58	58	73	73	73	73
	inch	2.3	2.3	2.3	2.9	2.9	2.9	2.9
K	mm	176	176	176	266	266	271	266
	inch	6.9	6.9	6.9	10.5	10.5	10.7	10.5
L	mm	276	276	276	451	482	482	451
	inch	10.9	10.9	10.9	17.8	19.0	19.0	17.8





ATS UUL ULTRA LOW PROFILE 0.5ton - 20ton

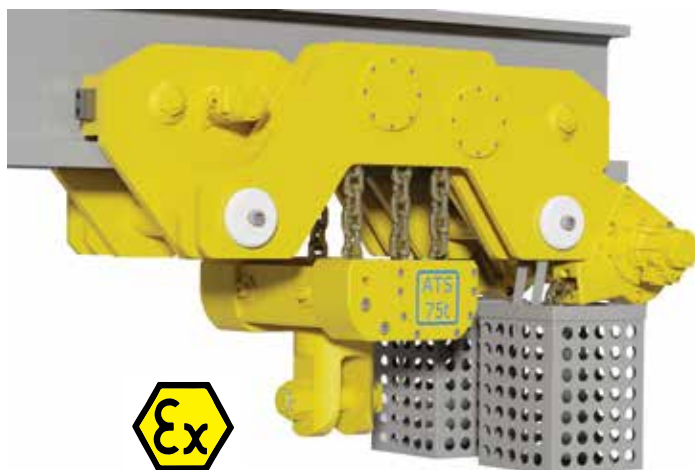
TECHNICAL DATA - ATS UUL ULTRA LOW PROFILE HOISTS - 0.5ton - 20ton

		UUL-05	UUL-1	UUL-2	UUL-3	UUL-4	UUL-6	UUL-10	UUL-12	UUL-16	UUL-20
Capacity	mt	0.5	1	2	3	4	6	10	12	16	20
	T	0.55	1.1	2.2	3.3	4.4	6.6	11.0	13.2	17.6	22.0
Unit Weight with 10 foot lift	lbs.	198	211	219	465	496	510	1221	1221	1307	1331
Weight of Chain per additional foot of lift	lbs. / ft.	2.2			6		12		24		48
Chain Size		7x21			9x27		13x36	16x45	16x45	13x36	16x45
Number of chain strands		2			2		2	2	2	4	
Minimum Headroom	mm	143			170		198	257	279	310	370
	inches	5.63			6.69		7.80	10.12	10.98	12.20	14.57
Air Pressure	bar	4-7									
Air Consumption	cfm	95			110						
Motor Power Output	kW	1.4						3.2			
	H.P.	1.9						4.3			
Lifting Speed at Full Load	ft./min	32	12	11	15	14	10	9	7	6	4.2
Lifting Speed without Load		61	18	18	21	21	14	9.5	9.5	8	9
Lowering Speed with Full Load		49	18	20	21	22	14	9.1	10	9.5	12
Traveling Speed	ft./min	19			39						
Air Hose Connection	inch	3/4					1			1.5	
Air Hose Dimension	inch	3/4					1				
Traversing I-Beam Width	inch	0.9-3.3			1.3-4.9						
Maximum Bottom Flange Thickness	inch	0.9			1.3						
Noise Level at Motorized Trolley	dB(A)	84									

* Specific Dimensions Provided Upon Request

** Increased speed conversion kits are available





ATS UUL ULTRA LOW PROFILE LARGE CAPACITY 50t - 120ton

TECHNICAL DATA - ATS UUL ULTRA LOW PROFILE LARGE CAPACITY - 50ton - 120ton

		UUL-50 - 60	UUL-75 - 90	UUL-100 - 120
Capacity	mt	50 - 60	75 - 90	100 - 120
	T	55 - 66	82.6 - 99	110 - 132
Unit Weight with 10 foot lift	lbs.	4850	7385	10119
Weight of Chain per additional foot of lift	lbs. / ft.	33	49	66
Chain Size		23.5x66		
Number of chain strands		4	6	8
Minimum Headroom	mm	442	520	720
	inches	17.4	20.5	28.3
Air Pressure	bar	5-7		
Air Consumption	cfm	320		
Motor Power Output	kW	5x2		
	H.P.	6.7x2		
Lifting Speed at Full Load	ft./min	3.6	2.1	1.6
Lifting Speed without Load		5.25	3.6	2.6
Lowering Speed with Full Load		6.2	3.9	2.9
Traveling Speed	ft./min	39		
Air Hose Connection	inch	1.5x2		
Air Hose Dimension	inch	1.5		
Noise Level at Motorized Trolley	dB(A)	84		

* Specific Dimensions Provided Upon Request



**CHESTER
HOIST**

By Columbus McKinnon

HOISTS & TROLLEYS

HAND CHAIN HOISTS

POWERED CHAIN HOISTS

TROLLEYS

END TRUCKS &
COMPONENTS



MODEL ALH **LOW-HEADROOM AIR CHAIN HOISTS**

CAPACITIES

1 to 24 Tons

LIFT

10 Feet Standard

Chester ALH Low Headroom Air Chain Hoists are ideal in manufacturing, power and industrial facilities, where precise load spotting is critical.

The standard pneumatic hoist package includes vane or piston motors and brakes with modulating valves for infinitely variable hoist motion. Single speed trolley is standard using vane motors and our standard worm gear reducers. Standard controls include a two or four lever pendant.

KEY FEATURES**ULTRA-LOW HEADROOM**

Design allows this hoist to operate in extreme low overhead locations where most other "low headroom" hoists can't.

COMPACT DESIGN

Integral hoist/trolley design provides compact overall envelope dimensions.

STRONG AND DURABLE

Steel construction to withstand harsh environments.

DIRECT-ACTING MOTOR BRAKES

Sets automatically if power is interrupted.

TWO PRECISION BEARINGS PER TROLLEY WHEEL

Minimizes traversing effort.

HOISTING CONTROL

Pilot air control permits variable speed hoisting and trolley traverse.

LIMIT SWITCH

Redundant limit switches for upper and lower hook travel.

TROLLEY

Plain, hand geared, or motor driven trolley models featuring anti-tilt rollers.

CUSTOM FIT

Can be made to fit virtually any beam.

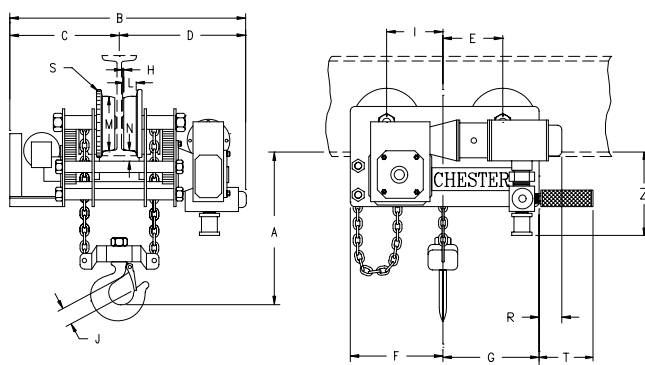
EASY INSTALLATION

Unique design allows easy installation on closed end monorail beam.

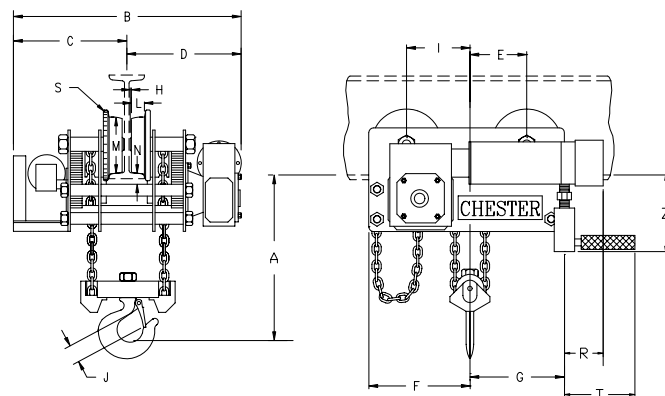


▲ **5-TON SPARK-RESISTANT MODEL
WITH MOTORIZED TROLLEY**

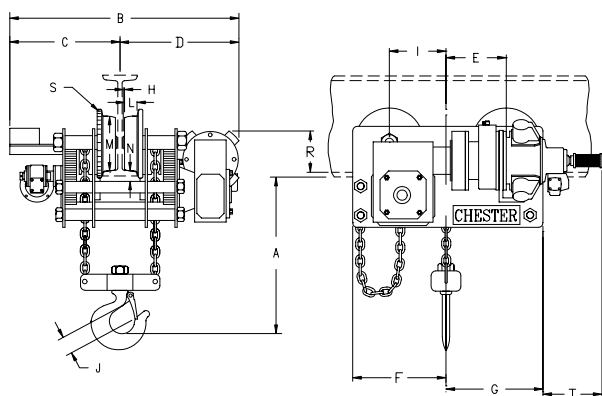
MODEL ALH LOW-HEADROOM AIR CHAIN HOISTS



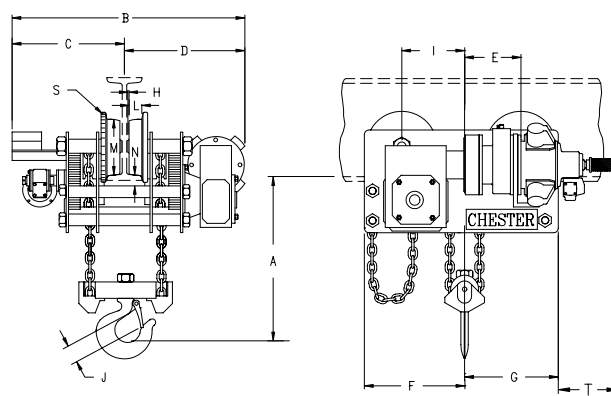
1 TO 2 TON
PLAIN, GEARED & MOTORIZED



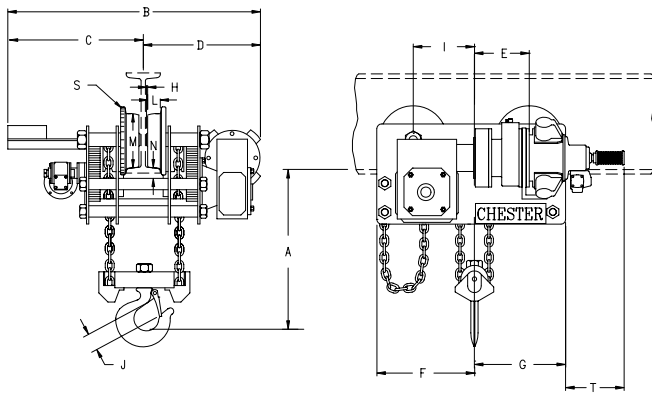
3 & 4 TON
PLAIN, GEARED & MOTORIZED



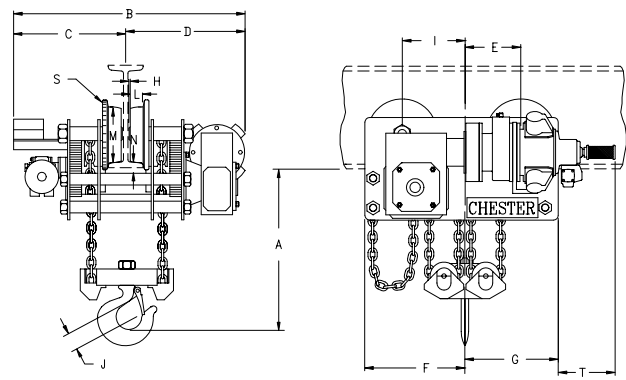
5 & 6 TON
PLAIN, GEARED & MOTORIZED



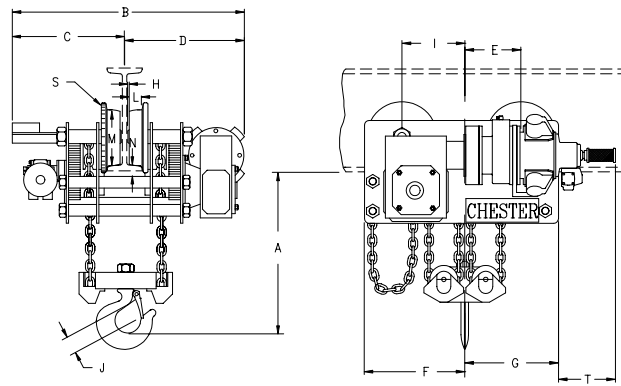
8 TON
PLAIN, GEARED & MOTORIZED



10 & 12 TON
PLAIN, GEARED & MOTORIZED



16 TON
PLAIN, GEARED & MOTORIZED



20 & 24 TON
PLAIN, GEARED & MOTORIZED

DIMENSIONS

Capacity (tons)	Dimensions (In.)																		
	A*	B	C	D	E	F	G	H**	I	J	L	M	N ***	P****	Beam Width	R	S	T*****	Z
1	7	35-5/8	16-1/2	19-1/8	7-3/4	10	10	1/16	7-3/4	1-1/8	1-1/8	4-1/2	7/8	S6x12.5#	3.332"	5-1/4	6	15	15
1-1/2	7	35-5/8	16-1/2	19-1/8	7-3/4	10	10	1/16	7-3/4	1-1/8	1-1/8	4-1/2	7/8	S6x12.5#	3.332"	5-1/4	6	15	15
2	7	35-5/8	16-1/2	19-1/8	7-3/4	10	10	1/16	7-3/4	1-1/8	1-1/8	4-1/2	7/8	S6x12.5#	3.332"	5-1/4	6	15	15
3	8	35-5/8	16-1/2	19-1/8	9	13	13	1/4	9	1-11/32	1-11/32	6-3/8	15/16	S8x18.4#	4.001"	5-3/16	8	10-1/2	12
4	8	35-5/8	16-1/2	19-1/8	9	13	13	1/4	9	1-11/16	1-11/32	6-3/8	15/16	S8x18.4#	4.001"	5-3/16	8	10-1/2	12
5	9-1/2	48--5/32	23-1/2	24-21/32	7-3/8	12	15	0	7-3/8	1-11/16	1-13/32	7-3/16	1-1/8	S10x25.4#	4.661"	—	9-3/8	20-7/16	—
6	9-1/2	48--5/32	23-1/2	24-21/32	7-3/8	12	15	0	7-3/8	1-11/16	1-13/32	7-3/16	1-1/8	S10x25.4#	4.661"	—	9-3/8	20-7/16	—
8	12	48--5/32	23-1/2	24-21/32	8-3/8	14	13-1/2	3/16	8-7/8	2-1/16	1-11/16	8-1/4	1	S10x25.4#	4.661"	—	10	20-11/16	—
10	12-1/2	48--5/32	23-1/2	24-21/32	8-1/2	15-5/8	14-3/8	3/8	9-3/4	2-1/4	1-3/4	9-3/4	1-1/8	S12x31.8#	5"	—	11-11/16	18-5/8	—
12	12-1/2	48--5/32	23-1/2	24-21/32	8-1/2	15-5/8	14-3/8	3/8	9-3/4	2-1/4	1-3/4	9-3/4	1-1/8	S12x31.8#	5"	—	11-11/16	18-5/8	—
16	14-1/2	53--7/32	25-1/2	27-23/32	11-7/8	18-5/8	18-5/8	3/8	11-7/8	3	2	11-3/4	1-1/8	S15x42.9#	5.501"	—	14-13/32	10-3/4	—
20	18	53--7/32	25-1/2	27-23/32	12-1/8	19-1/4	19	3/8	12-1/8	3-5/8	2	11-3/4	1-3/16	S18x54.7#	6.001"	—	14-13/32	11-1/8	—
24	18	53--7/32	25-1/2	27-23/32	12-1/8	19-1/4	19	3/8	12-1/8	3-5/8	2	11-3/4	1-3/16	S18x54.7#	6.001"	—	14-13/32	11-1/8	—

* To wheel-running surface of beam

** To under beam roller - closest point of contact

*** From face of the wheel to the end of the axle

**** Minimum standard I-beam for proper wheel running clearance only. Customers must verify their beam is adequate for the applied loading.

***** For standard motor and brake only. Hoists are designed for H-3 duty.

NOTE: must take flange width and thickness of beam into consideration.

Flat tread wheels (supplied for "W" beams) not shown.



Tiger

Product Catalogue

Tiger hoisting, clamping, winching, and height safety ranges

V1.1 | 2021 | tigerlifting.com

Air hoists



Tiger industrial air hoists are designed for, and proven in, many industries including mining, shipyards, offshore construction and power stations. They are also used for industrial and general engineering applications such as automotive plants, foundries, and heat treatment plants. The hoists operate at air pressures of between 4 and 6 bar. They comply with the requirements of ASME B30.16 and EN14492-2. Our manufacturing facility is ISO9001, ISO14001 & OHSAS18001 certified.

- Range - 250kg to 20 tonne
- Reliable, fast, efficient, quiet and lightweight
- Greater than 5:1 factor of safety
- Suitable for Fleeting/Oblique pulling
- Certified for ATEX Category 3 as standard. Category 2 available on request

TAHS



250kg, 500kg, 1t & 2t

TAH21



1.5t, 2t & 3t

TAH33



3t, 6t & 10t

TAH52



5t, 10t, 16t & 20t

Pendant control

Our air hoists are supplied with either aluminium hard anodised controls which offer excellent corrosion protection or our Nickel plated SG iron option (for higher impact resistance combined with greater tensile strength and superior corrosion protection). Twist grip pendants are available as an option. Our new dual control pendant will operate two hoists with one pendant control. We also have a new remote control option available.



Aluminium hard anodised pendant



Nickel plated SG iron pendant



Optional twist grip pendant



New dual hoist pendant



New remote control

New

Load limiter

Integral differential pressure load limiting valve automatically prevents the operator lifting a load of more than 125 % of the working load limit. Load limiter is 100 % repeatable at all working pressures and does not wear or generate damaging heat.

Emergency stop

Pendant control hoists are supplied with a mains emergency stop valve. This shuts off the main air supply to the hoist when activated.

Temperature Range

Ambient temperature range -20°C to +70°C (for temperatures outside of this range please contact us for details).

Precise control

Accurate and precise positioning of loads is standard for both toggle and pendent control models.

Fail safe disc brake

Non-asbestos self-adjusting disc brake is applied automatically when the air supply to the hoist is shut off. The brake is self-adjusting for the life of the brake and can hold more than 125 % of the working load limit.

Fleeting, cross hauling or oblique pulling

This type of application is when the load deviates from the vertical position with a force acting directly between load hook and suspension point. Our air hoists can be used for fleeting and oblique pulling operations with no de-ration of the hoist. Cross hauling can be done with a single hoist, two hoists operated independently or with two hoists operated off one dual hoist pendant control.

Integral silencer

The exhaust air silencer is built into the hoist gearbox housing and limits the noise level to below 85dB(A).

Load chain

Grade 8 load chain ensures stringent high-speed duty cycles can be met. The load chain is in accordance with the requirements of EN818-7 Appendix B.

Vane motor

Assisted vane lifting ensures maximum starting torque when starting under load.

Sealed gearbox

The sealed planetary gearbox ensures that the most stringent duty cycles can be met. The gears are heat treated and mounted on high strength, low friction bearings.

Ease of maintenance

The modular design of the hoist allows easy access to internal components such as the motor, gearbox, brake and load limiter. Common parts are used throughout the range.

Optional chain carrier

Designed to carry and hold the length of load chain required for the operation, our stainless steel chain bucket is available in different size options. These chain carriers prevent loose chain from getting snagged during lifting or lowering operations and protect any loose chain during transit to and from the operational site.

ATEX

Manufactured in accordance with ATEX Directive 2014/34/EU and Machinery Directive 2006/42/EC and suitable for ATEX Category 3 – normal safety. Our products are assigned as follows: ATEX marking: : Ex II 3 GD c IIB T4 IIIB T135°C. Suitable for gas and dust hazardous areas: Zone 2 (gas) and Zone 22 (dust).



Ex II 3 GD c IIB T4 IIIB T135°C

II	3	GD	c	IIB	T4	IIIB	T135°C	
								T135°C
								Dust Temperature Class: Maximum external surface temperature 135°C
								IIIB
								Groups of Dust: Protected for group IIIC which includes groups IIIA
								T4
								Gas Temperature Class: Maximum external surface temperature 135°C
								IIB
								Gas Explosion Group: Protected for group IIB which includes group IIA
								c
								Protection type: Design safety
								GD
								Ex Atmosphere: Gas and Dust
								3
								Category: Normal Safety
								II
								Equipment Group: surface industries

Also available for ATEX Category 2 – high safety – please contact us for more information.

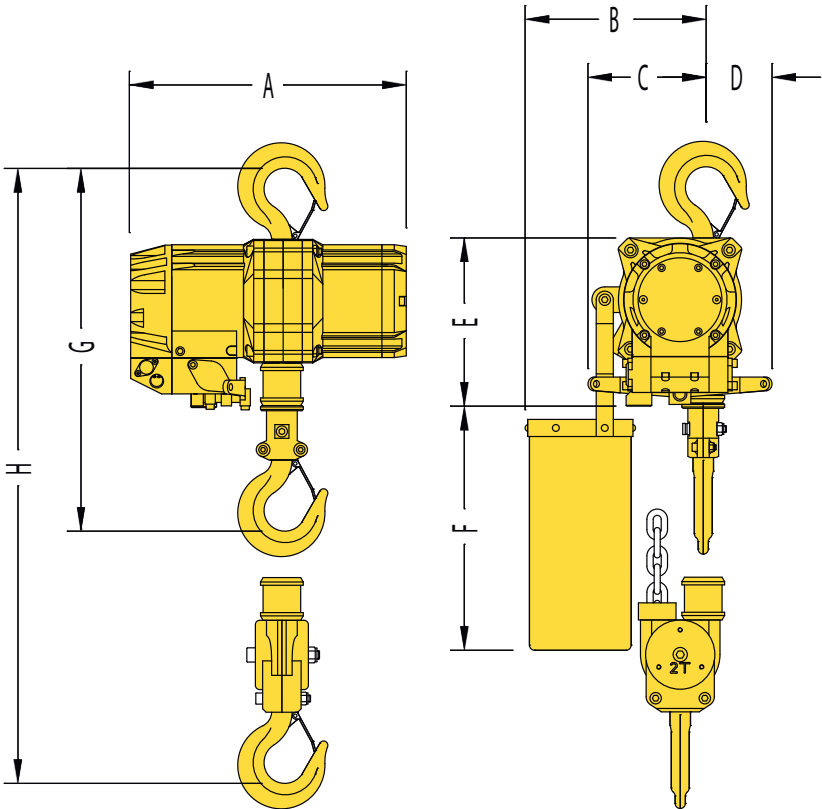
TAHS Range - 250kg, 500kg, 1.0, 2.0 tonne technical data (at 6 bar)

	AH-002	AH-005	AH-010	AH-020
Lifting Capacity (tonne)	0.25	0.5	1.0	2.0
Falls of Chain	1	1	1	2
Chain Size (mm)	7.1 x 21.2	7.1 x 21.2	7.1 x 21.2	7.1 x 21.2
Total Weight (kg)*	25.0	25.0	25.0	32.8
Full Load Lifting Speed (m/min)	22.5	15.0	7.5	3.8
No Load Lifting Speed (m/min)	32.0	29.5	16.0	8.0
Full Load Lowering Speed (m/min)	31.0	36.0	20.4	10.2
No Load Lowering Speed (m/min)	25.0	23.0	13.0	6.5
Air Consumption (m³/min)	2.1	2.1	2.1	2.1
Air Connection	G 1/2	G 1/2	G 1/2	G 1/2
Supply Hose (Inside ø) (mm)	16.0	16.0	16.0	16.0
Sound Level (dB(A))	76	76	76	76
Dimensions (mm): A	312	312	312	312
B	216	216	216	216
C	135	135	135	135
D	85	85	85	85
E	196	196	196	196
F (Min)	370	370	370	370
G (Min)	420	420	420	-
H (Min)	-	-	-	495

* Weight of TAHS model includes 3m lift, 2m toggle control. For longer lifts add 1.1kg for the AH-002, AH-005 and AH-010 models and 2.2kg for AH-020 per metre of lift. For pendant control add 0.8kg.



250kg TAHS with toggle control (AH-002T)



250kg-2t

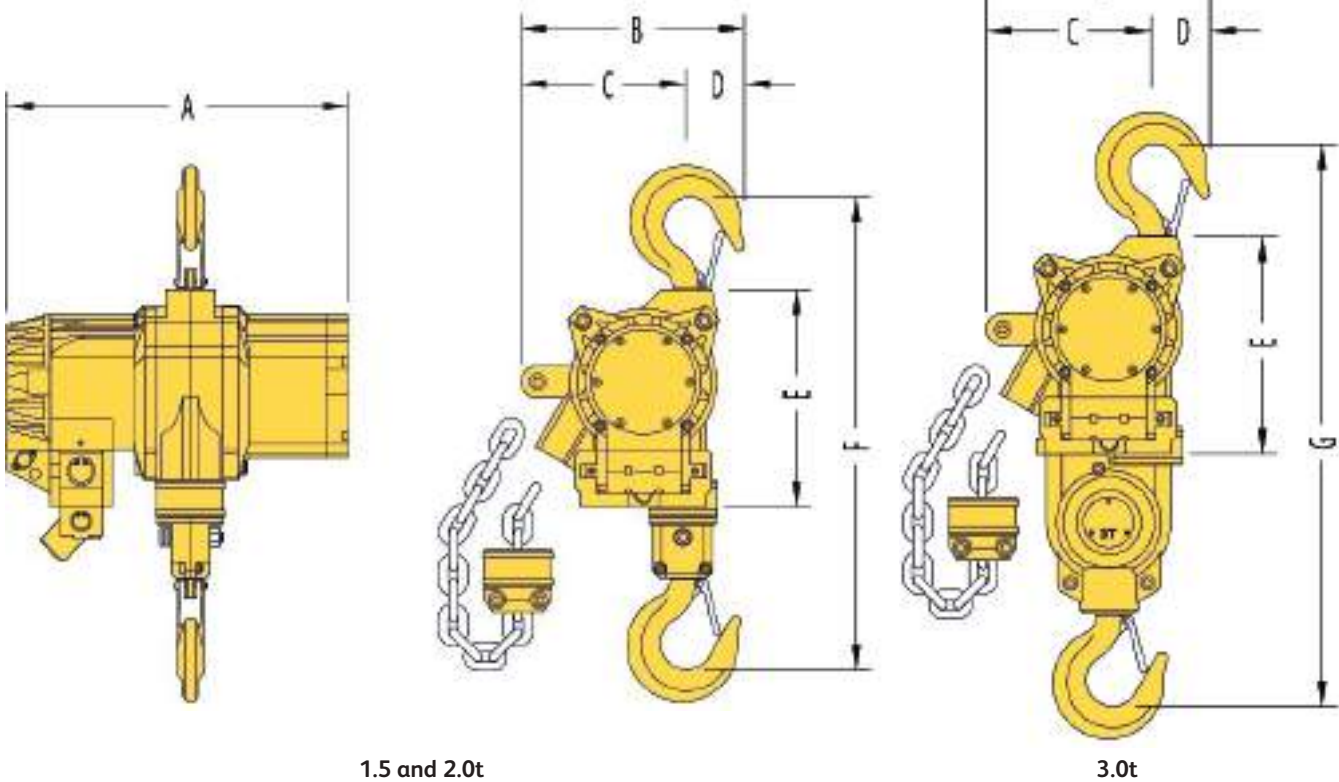
TAH21 Range - 1.5, 2.0, 3.0 tonne technical data

	AH-015	AH-021	AH-032
Lifting Capacity (tonne)	1.5	2.0	3.0
Falls of Chain	1	1	2
Chain Size (mm)	10 x 28	10 x 28	10 x 28
Total Weight (kg)*	55	55	67
Full Load Lifting Speed (m/min) @ 6 bar	6.2	5.3	3.1
Full Load lifting Speed (m/min) @ 4 bar	4.2	3.0	2.1
No Load Lifting Speed (m/min) @ 6 bar	8.7	8.7	4.3
Full Load Lowering Speed (m/min) @ 6 bar	8.6	9.0	4.3
No Load Lowering Speed (m/min) @ 6 bar	7.5	7.5	3.7
Air Consumption (m³/min) @ 6 bar	2.6	2.6	2.6
Air Consumption (m³/min) @ 4 bar	1.7	1.7	1.7
Air Connection	G 3/4	G 3/4	G 3/4
Supply Hose (Inside ø) (mm)	19	19	19
Sound Level (dB(A)) Full Load Lifting @ 6 bar	81	81	81
Full Load Lowering @ 6 bar	83	83	83
Dimensions (mm): A	382	382	382
B	250	250	250
C	184	184	184
D	66	66	66
E	213	213	213
F (Min)	515	515	-
G (Min)	-	-	610

* Weight of TAH21 model includes 3m lift, 2m toggle. For longer lifts add 2.2kg per metre of lift for the AH-015 and AH-021 and 4.4kg for the AH-032 per metre of lift. For pendant control add 0.4kg, for E/Stop pendant add 2.1kg.



2t TAH21 with E-stop pendant control (AH-021P)



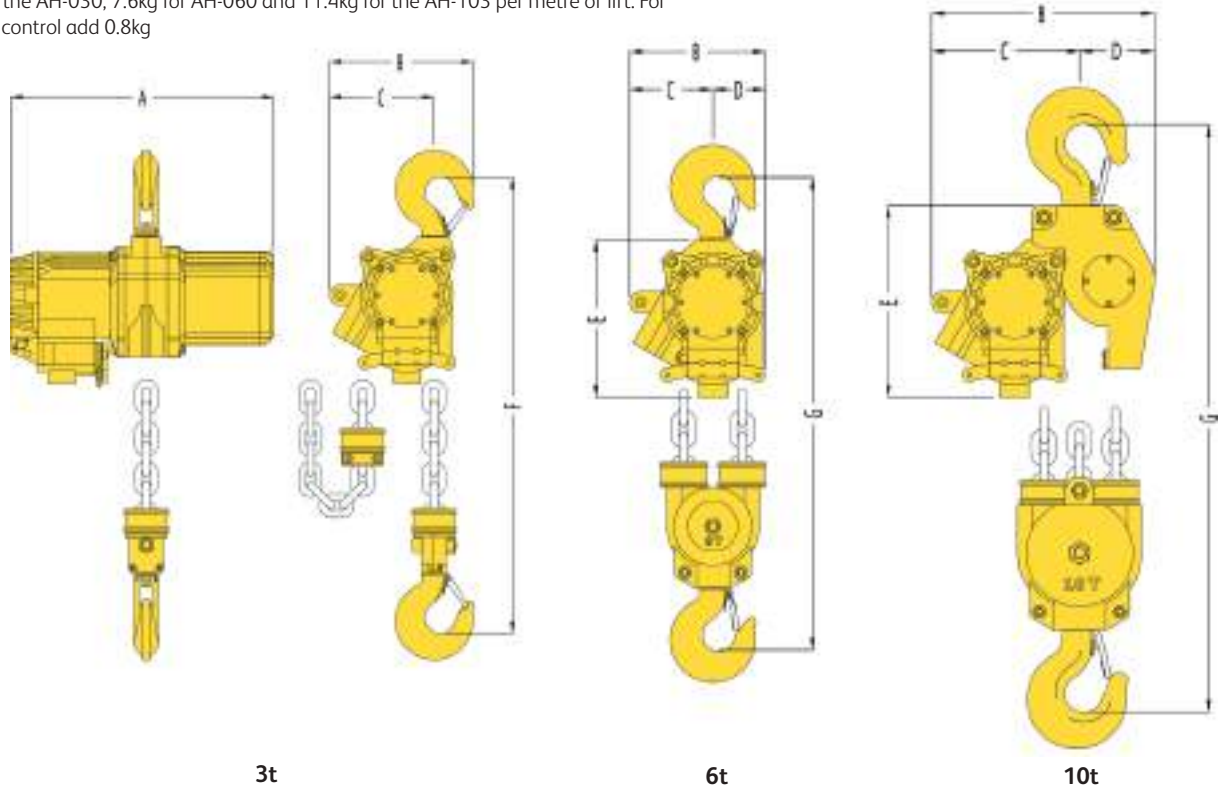
TAH33 Range - 3.0, 6.0, 10.0 tonne technical data

	AH-030	AH-060	AH-103
Lifting Capacity (tonne)	3.0	6.0	10.0
Falls of Chain	1	2	3
Chain Size (mm)	13 x 36	13 x 36	13 x 36
Total Weight (kg)*	76.5	99.5	139.5
Full Load Lifting Speed (m/min)@ 6 bar	5.3	2.6	1.6
Full Load Lifting Speed (m/min) @ 4 bar	3.8	1.9	1.1
No Load Lifting Speed (m/min) @ 6 bar	7.5	3.7	2.4
Full Load Lowering Speed (m/min) @ 6 bar	8.3	4.1	2.6
No Load Lowering Speed (m/min) @ 6 bar	6.8	3.4	2.2
Air Consumption (m³/min) @ 6 bar	4.0	4.0	4.0
Air Consumption (m³/min) @ 4 bar	2.6	2.6	2.6
Air Connection	G 3/4	G 3/4	G 3/4
Supply Hose (Inside ø) (mm)	25	25	25
Sound Level (dB (A)):			
Full Load Lifting @ 6 bar	81	81	81
Full Load Lowering @ 6 bar	83	83	83
Dimensions (mm): A	466	466	466
B	254	223	366
C	185	137	243
D	N/A	86	123
E	N/A	259	315
F (Min)	538	N/A	N/A
G (Min)	N/A	630	786



3t TAH33 with E-stop pendant control (AH-030P)

* Weight of TAH33 model includes 3m lift, 2m toggle. For longer lifts add 3.8kg per metre of lift for the AH-030, 7.6kg for AH-060 and 11.4kg for the AH-103 per metre of lift. For pendant control add 0.8kg



TAH52 Range - 5.0, 10.0, 16.0, 20.0 tonne technical data

	AH-050	AH-102	AH-160	AH-200
Lifting Capacity (tonne)	5.0	10.0	16.0	20.0
Falls of Chain	1	2	3	4
Chain Size (mm)	16 x 45	16 x 45	16 x 45	16 x 45
Total Weight (kg)*	106	152	245	288
Full Load Lifting Speed (m/min)@ 6 bar	3.0	1.5	0.9	0.7
Full Load Lifting Speed (m/min) @ 4 bar	1.9	0.9	0.6	0.4
No Load Lifting Speed (m/min) @ 6 bar	4.0	2.0	1.1	0.9
Full Load Lowering Speed (m/min) @ 6 bar	4.4	2.2	1.3	1.0
No Load Lowering Speed (m/min) @ 6 bar	3.3	1.7	1.0	0.8
Air Consumption (m³/min) @ 6 bar	4.0	4.0	4.0	4.0
Air Consumption (m³/min) @ 4 bar	2.6	2.6	2.6	2.6
Air Connection	G 3/4	G 3/4	G 3/4	G 3/4
Supply Hose (Inside ø) (mm)	25	25	25	25
Sound Level (dB(A)):				
Full Load Lifting @ 6 bar	82	82	82	82
Full Load Lowering @ 6 bar	83	83	83	83
Dimensions (mm): A	503	503	616	616
B	300	270	290	280
C	230	170	180	165
D	N/A	100	110	115
E	N/A	286	325	325
F (Min)	600	N/A	N/A	N/A
G (Min)	N/A	790	905	920

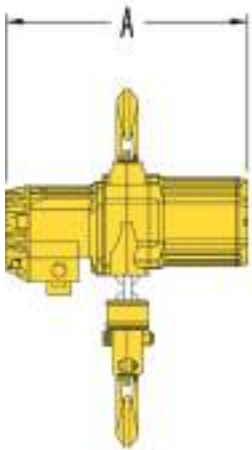


16t TAH52 with toggle control (AH-160T)

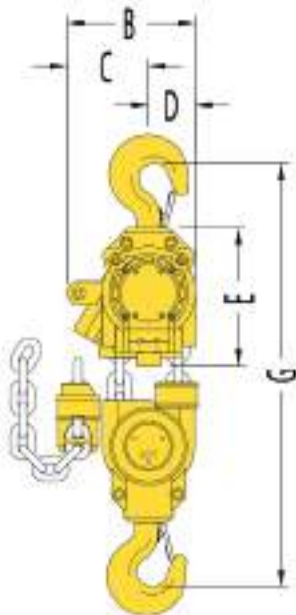
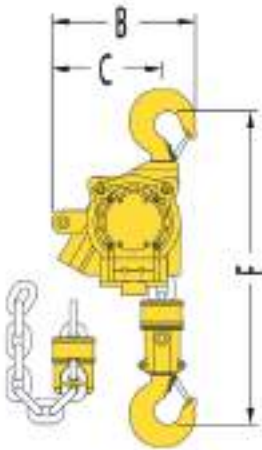
5t TAH52 with remote control (AH-050R)

* Weight of TAH52 model includes 3m lift, 2m toggle control. For longer lifts add 5.7kg for the AH-050, 11.4kg for the AH-102, 17.1kg for the AH-160 and 22.8kg for the AH-200 per metre of lift. For pendant control add 0.8kg

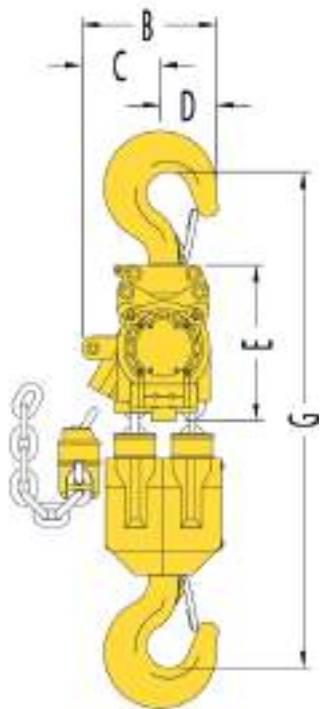
Due to our policy of continual product development, dimensions, weights and specifications may change without prior notice. Please check with your Tiger sales team when ordering.
Bespoke designs available on request- contact info@tigerlifting.com for more information.



5t



10t



16.0t/20.0t

Low headroom air trolley and hoist



ULT series hoists are designed specifically for mining, industrial and general engineering applications where headroom is extremely limited such as locomotive battery bays, pump chambers and underground workshops. They are available in geared and motorized traverse versions. The hoists operate at air pressures of between 4 and 6 bar and have a factor of safety in excess of 5 at the rated load. They comply with the requirements of SANS 1638, ASME B30.16 and EN14492-2. The manufacturing facility producing the hoists is ISO9001, ISO14001 & OHSAS18001 certified.

- Range - 2, 3, 4 & 6 tonne
- Ultra-low headroom, compact and versatile
- Available in geared and motorized versions
- Greater than 5:1 factor of safety
- Integrated load limiter

	ULTA2200	ULTA3200	ULTA4200	ULTA6200
Lifting Capacity (tonne)	2.0	3.0	4.0	6.0
Falls of Chain	2	2	2	4
Chain Size (mm)	10 x 28	10 x 28	10 x 28	10 x 28
Total Weight (kg)*	230	235	235	265
Full Load Lifting Speed (m/min) @ 6 bar	5.3	4.8	3.6	2.4
Full Load Lifting Speed (m/min) @ 4 bar	3.2	3.0	2.3	1.5
Traverse Speed (m/min) @ 6 bar	28.0	28.0	28.0	27.0
Air Consumption (m³/min) @ 6 bar	2.6	4.0	4.0	4.0
Air Consumption (m³/min) @ 4 bar	1.7	2.6	2.6	2.6
Air Connection	G 3/4	G1	G1	G1
Supply Hose (Inside ø) (mm)	19	25	25	25
Sound Level (dB(A))	83	83	83	83
Minimum Headroom (mm)	260	265	265	234
Minimum Beam Curve Radius (mm)	3600	3600	3600	4200

* Weight includes 3m lift, 2m pendant control.
* For longer lifts add 4.4 kg for the ULTA2200, ULTA3200, ULTA4200 and 8.8kg for the ULTA6200.

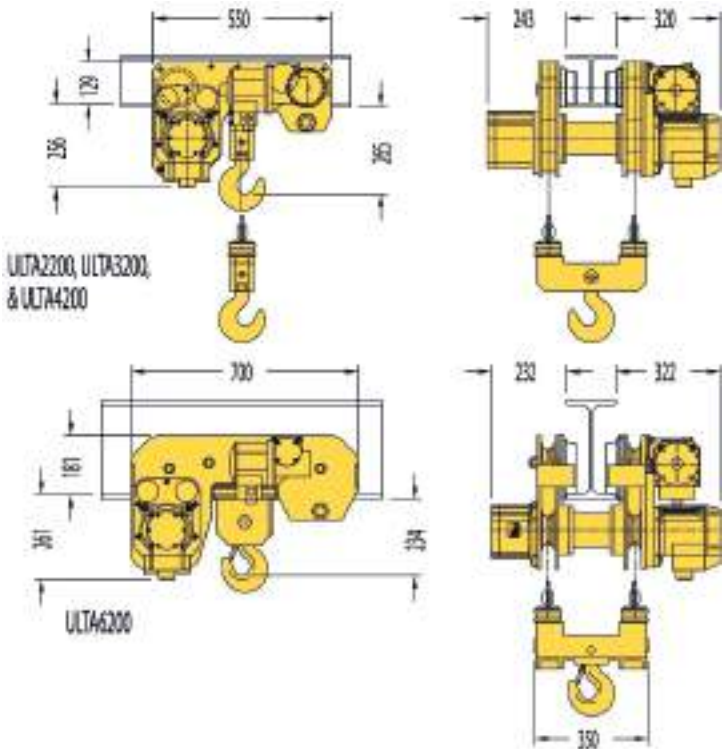
Monorail hoists

10t and 20t monorail hoists now available - ask us for details.



Bespoke designs available on request-
contact info@tigerlifting.com for more information.

Due to our policy of continual product development, dimensions, weights and specifications may change without prior notice. Please check with your Tiger sales team when ordering.

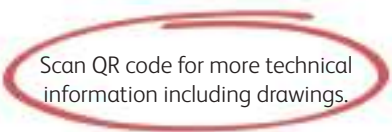


Air trolleys



T series trolleys are designed specifically for mining, industrial and general engineering applications. The trolleys are available in plain, geared and motorized versions and can be supplied with rigid or hook on adaptors. They have a minimum factor of safety of 5 at the maximum rated load and are therefore suitable for use with both manual and powered hoists. The trolleys comply with the requirements of SANS 1824, EN13157 and EN1449-2. The manufacturing facility producing the trolleys is ISO9001, ISO14001 & OHSAS18001 certified.

- 1, 3 and 6 tonne capacity units
- Low headroom, compact and versatile
- Available in plain, geared and motorized versions
- Greater than 5:1 factor of safety
- Anti-fall, anti-derailment and anti-tilt features as standard



HOISTING

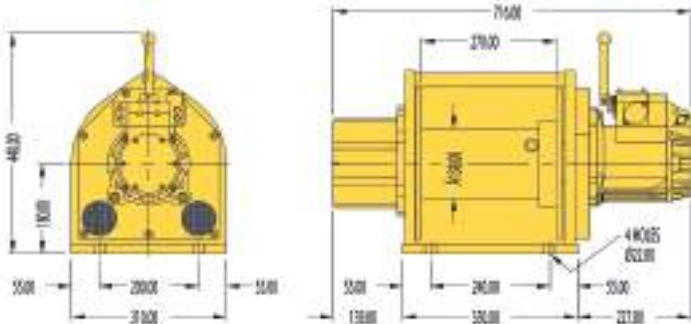
Product Code	Capacity (tonne)	Track width range (mm) A	Min. curve ratio (m)	Net Weight (kg)	Max. Traverse Speed (m/min) @ 6 bar
TA1000	1.0	73-152	900	30.5	22.0*
TG1000	1.0	73-152	900	21.2	-
TP1000	1.0	73-152	900	18.2	-
TA3000	3.0	89-184	950	44.0	28.0*
TG3000	3.0	89-184	950	34.7	-
TP3000	3.0	89-184	950	31.7	-
TA6000	6.0	128-184	1600	80.0	27.0*
TG6000	6.0	128-184	1600	72.0	-
TP6000	6.0	128-184	1600	69.0	-

Air winch



- Range - 0.5, 1, 2 & 3 tonne
- Reliable, fast efficient and quiet
- Compact design
- Extended drum versions available for increased rope capacity
- Easy to mount
- Greater than 5:1 factor of safety

Technical Sepecifications at 6 bar	WAR500I	WAR1000I	WAR2000I	WAR3000I
Half Drum Line Pull (kg)	500	1000	2000	3000
Rope Speed at Rated Load (m/min)	28.10	13.90	6.70	4.60
Full Drum No Load Speed (m/min)	60.30	28.20	18.50	9.00
Air Consumption (m³/min)	4.0	4.0	4.0	4.0
Air Connection	G 3/4	G 3/4	G 3/4	G 3/4
Supply Hose (Inside ø) (mm)	25	25	25	25
Sound Level (dB(A))	80	80	80	80
Reccommended Rope Size (mm)	13.0	13.0	15.0	18.0
Rope Storage Length (m)- Standard Drum	80	80	59	52
Rope Storage Length (m)- Extended Drum	165	165	119	106
Weight without Wire Rope (kg) - Std	104	104	104	104





Specifications for Hoists and Trolleys



Lodestar Air XL Pneumatic Chain Hoist

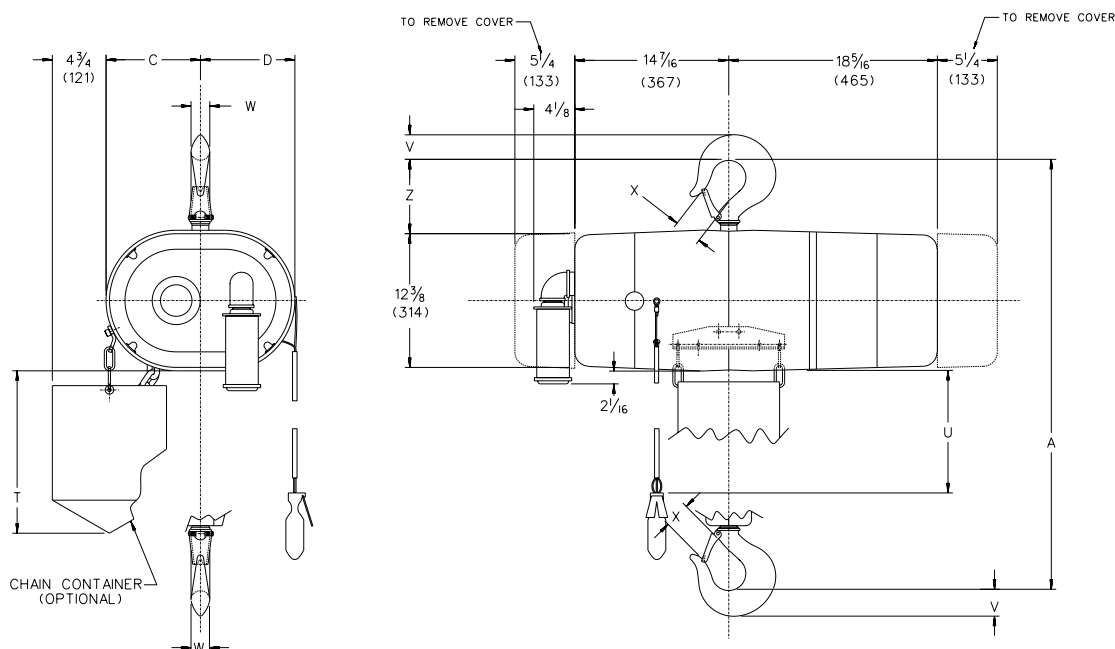


Lodestar Air XL chain hoists with capabilities from 2 to 7¹/₂ tons provides heavy-duty hoists featuring variable speeds and higher capacities than traditional air chain hoists. Workers in automotive plants, heavy equipment manufacturing, chemical plants, refineries, and related rugged work environments will experience the same dependability and versatility that they have relied on from CM products for many years.

The features of the Lodestar Air XL Chain Hoist includes:

- Lifting capacities up to 7¹/₂ tons
- Variable lifting speeds
- Maximum lifting speed of 31 feet per minute
- Forged ten-pocket oblique-lay liftwheel for smooth chain operation, constant chain speed, and reduced chain wear
- Standard overload protector
- Clutch acts as upper and lower limits
- Six-vane rotary air motor
- Pendant controls provide precise load spotting
- Air actuated motor brake for positive load control
- Factory tested at 125% of rated capacity
- Spark resistant models available
- Lifetime warranty
- Made in U.S.A.
- Standard Lift: 10 ft.
- Air inlet size: 1" NPT
- Air supply hose size: 1" I.D. min.
- Consumption: 175 CFM at 90 PSI
- Exhaust: 1 1/4" NPT

Lodestar Air XL with Hook Suspension-Specifications



Clearance Dimensions (inches)**

Dimensions	Single Reeved 2 ton capacity	Double Reeved 3-5 ton capacity	Triple Reeved 6 ton capacity
A	25	35 5/16	36 3/16
C	8 7/16	11 3/8	11 3/8
D	8 1/8	5 1/2	5 1/2
T (based on 10 ft. lift)	14 3/16	14 3/16	14 3/16
U (based on 10 ft. lift)	6'- 3"	6'- 3"	6'- 3"
V	1 1/2	2 1/4	2 1/4
W	1 1/8	1 5/8	1 5/8
X	1 5/16	2 1/16	2 1/16
Z	4 1/8	5 1/2	5 1/2
Weight (lbs.)	347	421	453
Lifting Speed (FPM)	31	21/13	10

**Dimensions listed are for minimum size "S" beam.

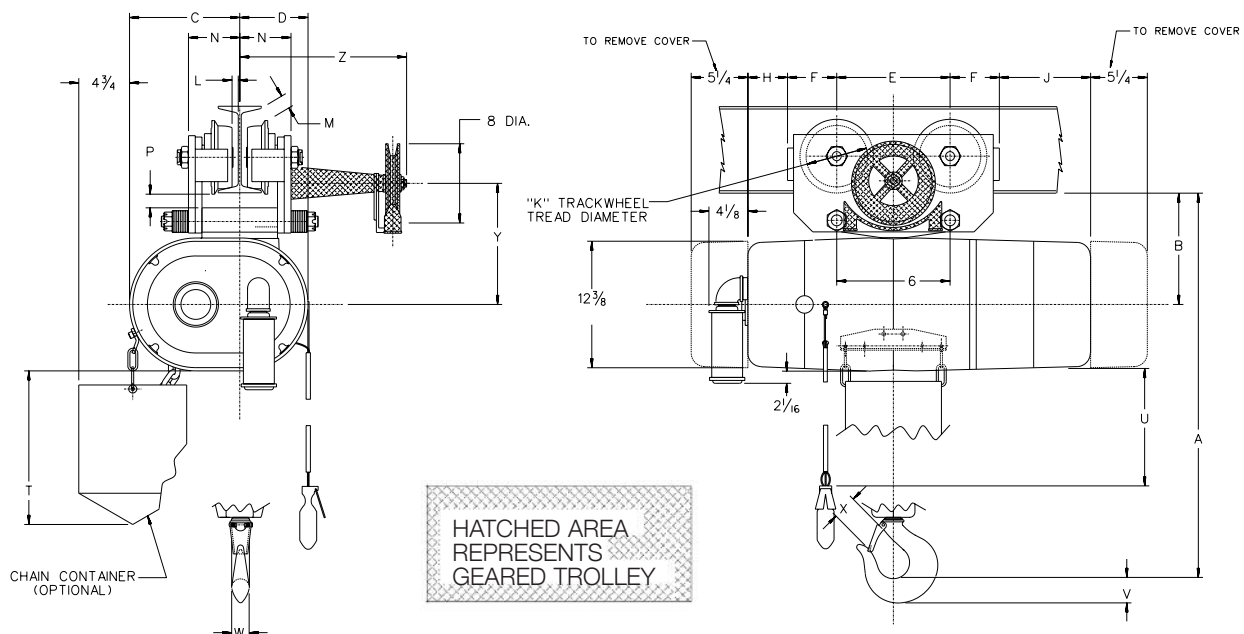
CM® Repair/Replacement Policy

All Hoists are thoroughly inspected and performance tested prior to shipment. If any properly maintained CM Hoist develops a performance problem due to a material or workmanship defect, as verified by Columbus McKinnon, repair or replacement of the hoist will be made to the original purchaser without charge. This repair/replacement policy applies only to Hoists installed, maintained and operated as outlined in the applicable accompanying CM Hoist Manual, and specifically excludes normal wear, abuse, improper installation, improper or inadequate maintenance, hostile environmental effects, and unauthorized repairs/modifications.

ANSI/OSHA

CM hoists are manufactured in accordance with our interpretation of the ASME/ANSI Code B30.16 for Overhead Hoists and U.S. Occupational Safety and Health Act regulations at the time of manufacture. Users should check each installation for compliance with the application, operation and maintenance requirements of these regulations and standards.

Lodestar Air XL with plain trolley and geared trolley-Specifications

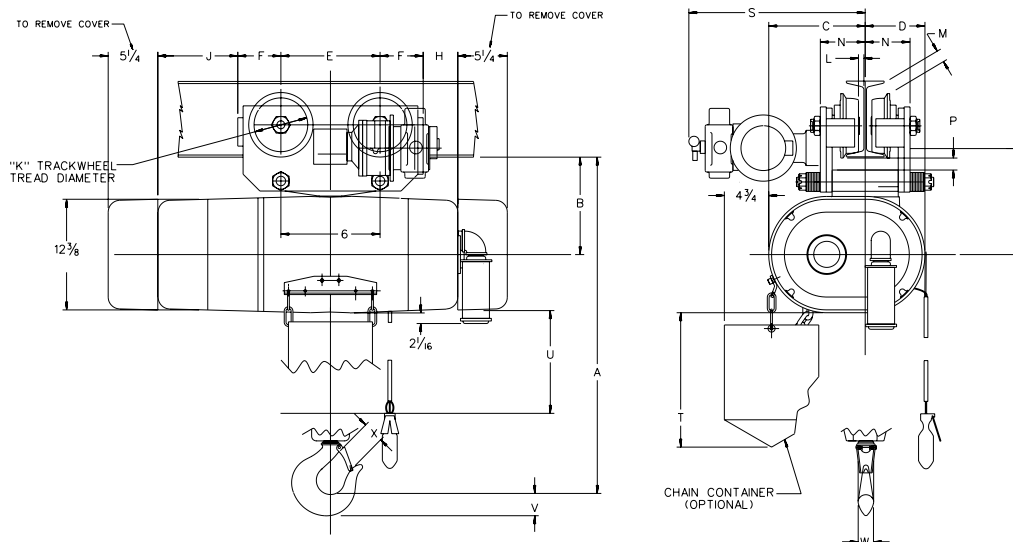


Dimensions	Clearance Dimensions (inches)**			
	Single Reeved 2 ton capacity	Double Reeved		Triple Reeved 6 & 7 1/2 ton capacity
A	28 11/16	33 1/2	33 3/4	34 5/8
B	9 3/4	10 7/16	10 11/16	10 1 1/16
C	8 7/16	11 3/8	11 3/8	11 3/8
D	8 3/8	5 1/2	5 1/2	5 1/2
E	9 1/2	9 1/8	12	12
F	3 1/2	3 1/4	4 13/16	4 13/16
H	5 1/4	5 11/16	2 11/16	2 11/16
J	10 1/16	10 1/2	7 1/2	7 1/2
K	4 11/16	4	6 1/2	6 1/2
L	1/2	5/8	5/8	5/8
M	1/2	2 11/16	0	0
N	5	3 5/8	3 7/8	3 7/8
P	1 1/4	3/8	7/16	7/16
T (based on 10 ft. lift)	14 3/16	14 3/16	14 3/16	14 3/16
U (based on 10 ft. lift)	6'- 3"	6'- 3"	6'- 3"	6'- 3"
V	1 1/2	2 1/4	2 1/4	2 1/4
W	1 1/8	1 5/8	1 5/8	1 5/8
X	1 5/16	2 1/16	2 1/16	2 1/16
Y*	11 5/16	12 7/8	11 1/8	11 1/8
Z*	—	19 1/8	19 1/8	19 1/8
Weight (lbs.)-Plain Trolley	407	476	576	608
Weight (lbs.)-Geared Trolley	452	491	586	618
Minimum Radius Curve (ft.)	4	4	6	6
Standard Beam Flange Adj. Range (in.)	3.33-5.55	4.00-6.25	4.00-6.25	4.00-6.25
Lifting Speed (FPM)	31	21	13	10/9

*Geared Trolley Dimensions Only

**Dimensions listed are for minimum size "S" beam.

Lodestar Air XL with Motor Driven Trolley-Specifications



Clearance Dimensions (inches)**

Dimensions	2 ton capacity	Double Reeved		Triple Reeved
		3 ton capacity	5 ton capacity	6 & 7½ ton capacity
A	33½	33½	33¾	34⅝
B	10⅞	10⅞	10⅞	10⅞
C	9⅝	11⅜	11⅜	11⅜
D	7⅞	5½	5½	5½
E	9⅞	9⅞	12	12
F	3¼	3¼	4⅜	4⅜
H	5⅞	5⅞	2⅞	2⅞
J	10½	10½	7½	7½
K	4	4	6½	6½
L	⅝	⅝	⅝	⅝
M	2⅞	2⅞	0	0
N	3⅝	3⅝	3⅞	3⅞
P	⅜	⅜	⅞	⅞
S	15⅞	15⅞	17¾	17¾
T (based on 10 ft. lift)	14⅞	14⅞	14⅞	14⅞
U (based on 10 ft. lift)	6'- 3"	6'- 3"	6'- 3"	6'- 3"
V	2¼	2¼	2¼	2¼
W	1⅝	1⅝	1⅝	1⅝
X	2⅞	2⅞	2⅞	2⅞
Minimum Radius Curve (ft.)	4	4	6	6
Weight (lbs.)	505	569	611	643
Standard Beam Flange Adj. Range (in.)	4.00-6.25	4.00-6.25	4.00-6.25	4.00-6.25
Trolley Speed (ft./min)	50	50	50	50
Standard Single Speed	50	50	50	50
Lifting Speeds	31	21	13	10/9

**Dimensions listed are for minimum size "S" beam.

⚠ WARNING

Failure to properly install, operate and maintain the Lodestar Air XL Hoist may result in injury and/or property damage.

TO AVOID INJURY:

- Installation, maintenance and operation must be carried out as instructed in the manual provided with the Lodestar Air XL Hoist.
- Do not use the Lodestar Air XL Hoist for lifting people; special equipment is required for this purpose. Refer to ASME/ANSI A-17 Series and applicable state and local codes for specific requirements.

ShopAir™ chain hoist

**Compact, powerful, and precise
for assembly and manufacturing
applications up to 1,000 pounds.**

The CM ShopAir combines superior lifting speeds, rugged design, and precise load spotting in a portable air chain hoist. As a workstation hoist or a production line hoist, the ShopAir is the ideal choice.

CAPACITIES: 250 to 1,000 Lbs.

**NORMAL
OPERATING
PRESSURE:** 90 PSI (6.2 bar)

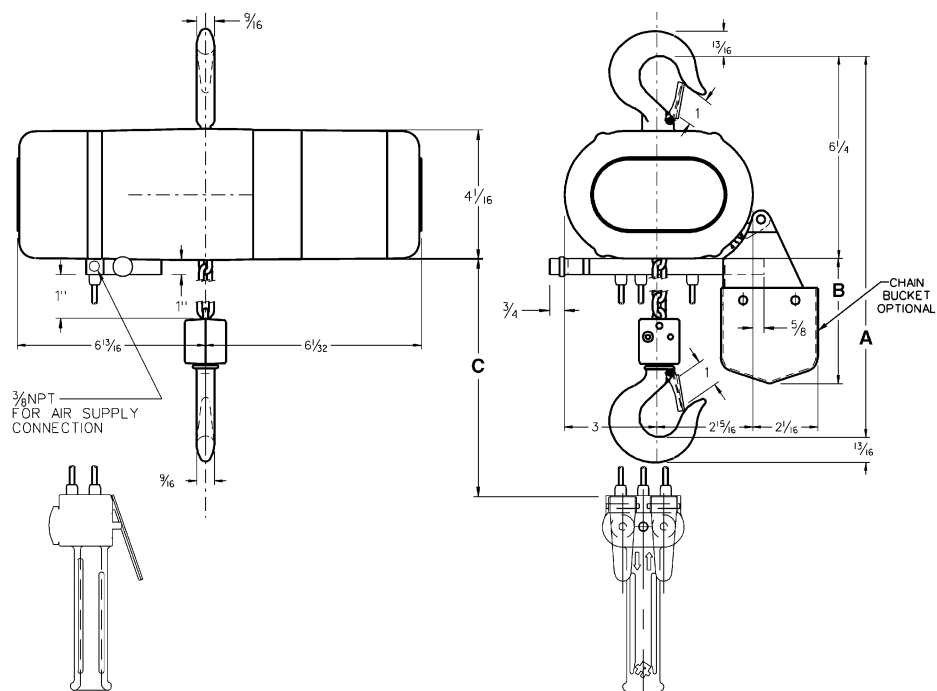
**AIR
CONSUMPTION:** 34 cu.feet/min.

- Standard Protector™ overload device.
- 10-pocket, oblique lay liftwheel provides longer chain life.
- Epoxy powder coat finish provides a durable exterior.
- Durable reduction gearing means reliable operation
- Variable flow, two lever pendant for precise load spotting.
- Threaded external exhaust for piping away exhaust in clean room or painting applications.
- Small, compact design for commercial & industrial applications.
- Industrial duty air motor for tough applications.
- Gear train is lifetime lubricated with non-oxidizing grease.
- Hardened forged steel, latch-type lower hook rotates 360°.
- Hardened forged steel, rigid latch-type upper hook.
- Rugged cast aluminum alloy hoist frame.
- Alloy load chain.
- Maximum pendant drop - 36 ft.
- Lifetime warranty against defects in materials and workmanship.
- Made in U.S.A.





ShopAir™ air chain hoist



Specifications

Product code	Cap. (lbs.)	(in.)			Number of load chains	Speed FPM (under full load)		Shipping weight (lbs.)
		A	B	*C		lifting	lowering	
2178	250	10 ^{7/8}	3 ^{7/8}	6' 4"	1	31	83	30
2180	300	10 ^{7/8}	3 ^{7/8}	6' 4"	1	31	83	30
2182	500	10 ^{7/8}	3 ^{7/8}	6' 4"	1	21	75	30
2179	500	11 ^{15/16}	5 ^{3/4}	6' 4"	2	16	42	34
2181	600	11 ^{15/16}	5 ^{3/4}	6' 4"	2	16	42	34
2183	1,000	11 ^{15/16}	5 ^{3/4}	6' 4"	2	11	38	34

300 Lb. model shown with optional chain container

***WARNING**

Overloading and improper use can result in injury.

To avoid injury:

- Do not exceed working load limit, load rating capacity
- Do not use to lift people or loads over people.
- use only alloy chain for overhead lifting.
- Read and follow all instructions.

AirStar chain hoist

The CM AirStar now boasts a longer service life and superior air leak protection. Design enhancements, including upgrades to the control manifold and air seals, give the AirStar best-in-class performance even for the most demanding applications.

With capacities from 1/4 to 1 Ton, this variable speed air chain hoist is compact, lightweight and ideal for heavy duty industrial and workstation duties.

CAPACITIES & LIFT

Capacities from 1/4 to 1 Ton. Metric Rated. 10 feet standard lift. Other lifts available.

LOW OPERATING COSTS

Air consumption rate is just 48 CFM, which means lower operating costs.

FAST LIFTING SPEEDS

As fast as 45 FPM lifting and 120 FPM lowering on the 1/2 Ton unit.

PORTABLE & RUGGED

Aluminum frame and end-cover makes it lightweight and a rugged design.

BUILT TO LAST

Hoistaloy® load chain for heavy duty, or stainless steel chain for spark resistant and medium duty loads.

SECURE & RELIABLE

Positive action, heavy-duty shoe type brake that stops and holds the load. Limit stops prevent over-travel when lifting and lowering.

EASY TO HANDLE

Inlet air swivel with built-in strainer provides free hoist movement.

EXTERNAL BRAKE ADJUSTMENT

Allows for quick adjustment of brake responsiveness and sensitivity for exceptional load positioning and control.

SMOOTH OPERATION

Multi-vane rotary air motor for high torque and smooth operation.

Equa-torque gearing (heat treated alloy steel spur gears) are accurately machined to provide a close meshed, compact gear reduction.

PENDANT THROTTLE OR PULL CORD CONTROL

Ergonomic pendant throttle control features fast pendant-to-hook reaction time and one-handed operation.

SPECIAL APPLICATIONS

Tapped exhaust port (1/2 NPTF) for cleanroom applications or add a supplemental muffler for even quieter operation.

METAL CHAIN CONTAINER STANDARD

FLEXIBILITY

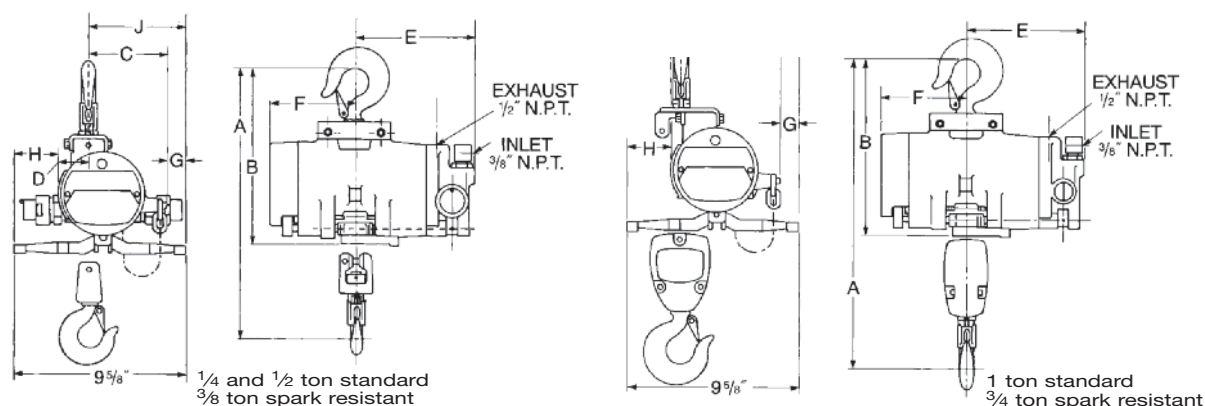
Hook or lug suspension available.

SPARK-RESISTANT

Spark resistant models available for hazardous environments.



AirStar with hook suspension-Specifications



Clearance dimensions (in.)

Rated capacity* (tons)	Product code	A	B	C	D	E	F	G	H	J
1/4	7400A	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/4	7400B	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/4	7400C	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/4	7400D	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401A	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401B	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401C	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401D	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
3/8	7403A	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
3/8	7403B	15 1/4	10 1/4	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1	7402A	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
1	7402B	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
1	7402C	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
1	7402D	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
3/4	7404A	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
3/4	7404B	18	10 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16

Specifications

Type	Rated capacity* (tons)	Pull cord control Product code no.	Net wt. (lbs.)	Pendant throttle control** Product code no.	Net wt. (lbs.)	Reeving	Lifting speed max.† (F.P.M.)	Lowering speed max.† (F.P.M.)
Standard models								
Roller chain	1/4	7400A	37	7400B	47	1	65	95
	1/2	7401A	37	7401B	47	1	45	120
	1	7402A	51	7402B	61	2	23	60
Link chain	1/4	7400C	36	7400D	46	1	65	95
	1/2	7401C	36	7401D	46	1	45	120
	1	7402C	49	7402D	59	2	23	60
Spark resistant models								
Link chain	3/8	7403A	36	7403B	46	1	60	100
	3/4	7404A	49	7404B	59	2	29	52

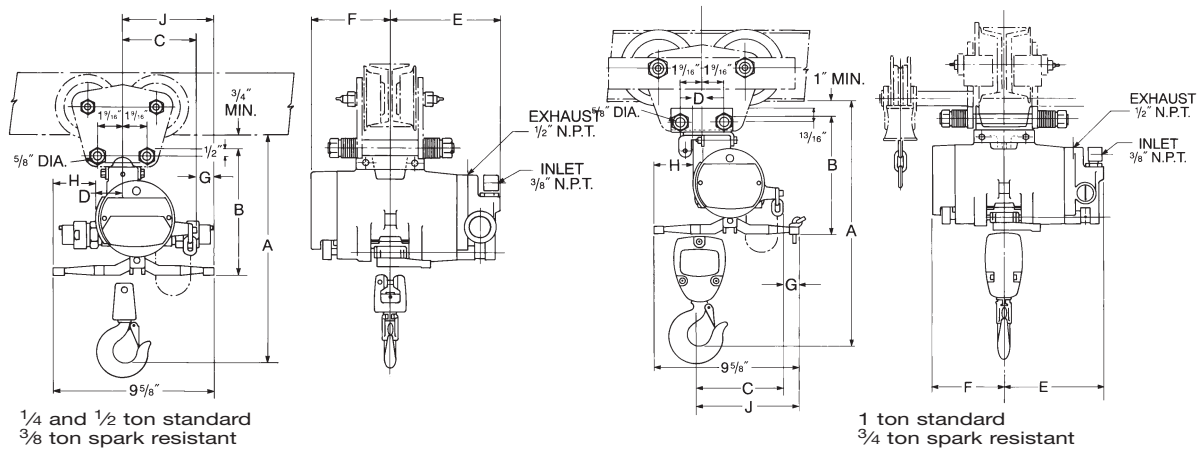
**Equipped with 6 ft. pendant control.

† Speeds based on 90 PSI line pressure and with full rated hoist load.

Hoist speeds are adjustable and can be reduced to 50% of full rated speed. Shipped from factory with hoisting speed set at maximum rated speed; lowering speed set at reduced speed.

Standard lift 10 feet. Longer lifts available.

AirStar with lug suspension-Specifications



Clearance dimensions (in.)

Rated capacity* (tons)	Product code	A	B	C	D	E	F	G	H	J
1/4	7400F	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/4	7400G	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401F	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1/2	7401G	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
3/8	7403C	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
3/8	7403D	13 7/8	8 1/8	4 3/4	1 3/4	7 1/16	5 1/16	15/16	2 3/16	5 11/16
1	7402F	17	8 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
1	7402G	17	8 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
3/4	7404C	17	8 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16
3/4	7404D	17	8 1/4	5 3/4	3/4	7 1/16	5 1/16	15/16	2 3/16	6 11/16

Specifications

Type	Rated capacity* (tons)	Pull cord control		Pendant throttle control**		Reeving	Lifting speed max.† (F.P.M.)	Lowering speed max.† (F.P.M.)
		Product code no.	Net wt. (lbs.)	Product code no.	Net wt. (lbs.)			
Standard models								
Link chain	1/4	7400F	36	7400G	46	1	65	95
	1/2	7401F	36	7401G	46	1	45	120
	1	7402F	49	7402G	59	2	23	60
Spark resistant models								
Link chain	3/8	7403C	36	7403D	46	1	60	100
	3/4	7404C	49	7404D	59	2	29	52

**Equipped with 6 ft. pendant control.

† Speeds based on 90 PSI line pressure and with full rated hoist load.

Hoist speeds are adjustable and can be reduced to 50% of full rated speed. Shipped from factory with hoisting speed set at maximum rated speed; lowering speed set at reduced speed.

Standard lift 10 feet. Longer lifts available.

Spark Resistant Models

Spark resistant models offer maximum protection in hazardous environments. Components for spark resistance include bronze alloy hooks, stainless steel hook latches, non-magnetic chrome-nickel stainless steel load chain, modified load sprocket, guide and stripper arrangement and lower blocks of stainless steel for the 3/8-ton single line model and anodized aluminum body on the 3/4-ton two part reeved model.

The spark resistant models have capacities derated. The stainless steel load chain is lower strength and not as hard as the standard alloy steel load chain, thereby requiring reduced capacity ratings on these models.

Warning: The equipment illustrated here is not designed or suitable as a power source for lifting or lowering persons. Exceeding rated capacity is dangerous and can result in injury to persons and damage property.

AirStar 6 air hoist

Designed for general industrial applications where variable speed pneumatic power is preferable. Rated for loads of up to 3 tons. This compact, lightweight hoist features:

- Hoist alloy load chain for heavy duty or stainless steel chain for spark resistant, medium duty loads
- Lightweight aluminum frame and cover
- Positive action, spring biased Weston-type load brake holds the load, provides exacting load spotting and control
- Multi-vane rotary air motor with eight vanes for high torque, smooth operation, positive starting
- Precision cut heat treated helical and spur gear combination for quiet, efficient operation
- Pendant throttle handle is full flow design for precision throttle capabilities
- Spark resistant models available for hazardous environments
- Hook or lug suspension available
- Standard overload protector device
- Internal Muffler—reduces sound level
- Throttle Valve—precision throttle valves meter the air for smooth operation
- Chain container optional
- Lifetime warranty
- Metric rated
- Made in U.S.A.

Basic Hoist Data

Rated Loads: 1, 2, and 3-ton (Spark resistant models are rated at 1/2, 1 and 2 ton)

Air Pressure

Recommended: 90 PSI

Air Consumption: 70 SCFM (full load-full speed) slow speed model 80 SCFM (full load-full speed) fast speed model

Pendant Control Hose: 6 foot pendant hose standard for 10 foot lift hoist, longer hose assemblies up to 36 foot available.

Lift: 10 foot lifts are standard. Longer lifts are an optional extra.

Net Wt. (Basic Hoist): 100 pounds

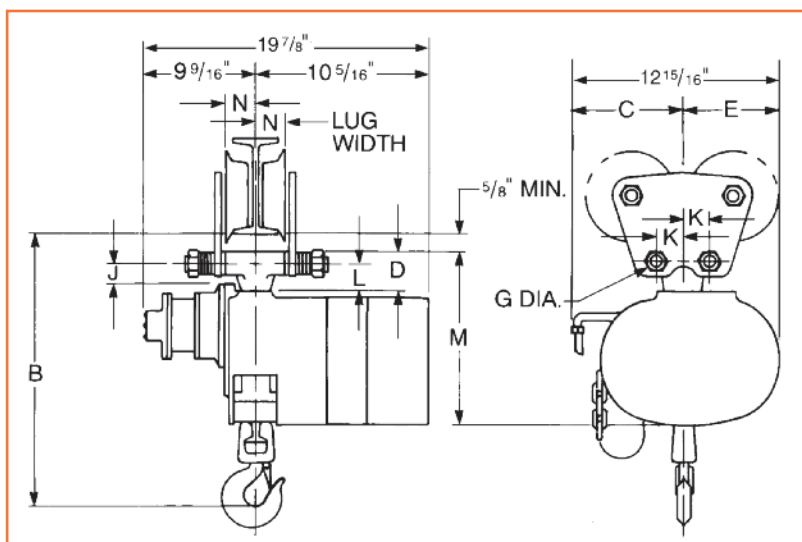
Suspension: Hook or lug

Air Inlet Size: 3/4 NPTF

Air Supply Hose Size: 3/4 I.D. min.



AirStar 6 with lug suspension



Clearance dimensions (in.)

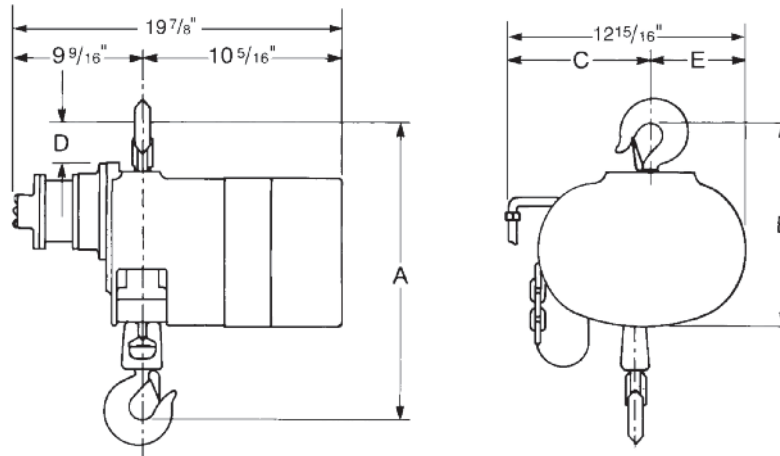
Rated capacity* (tons)	Product code	B	C	D	E	J	K	L	M	N	G
1/2	7408B	16 1/4	6 3/4	2	6 3/16	7/8	1 9/16	1 5/16	9 15/16	1 13/16	5/8
1	7405C	16 1/16	6 3/4	2	6 3/16	7/8	1 9/16	1 5/16	9 15/16	1 13/16	5/8
1	7405D	16 1/16	6 3/4	2	6 3/16	7/8	1 9/16	1 5/16	9 15/16	1 13/16	5/8
1	7409B	21 3/8	8 1/16	2 13/16	4 7/8	1 1/2	2 1/2	1 15/16	10 11/16	2 3/16	1
2	7406C	19 3/4	8 1/16	2 13/16	4 7/8	1 1/2	2 1/2	1 15/16	10 11/16	2 3/16	1
2	7406D	19 3/4	8 1/16	2 13/16	4 7/8	1 1/2	2 1/2	1 15/16	10 11/16	2 3/16	1
2	7410B	25	5 13/16	4 5/8	7 1/8	2 15/16	3	3 3/8	12 7/8	3	1 1/4
3	7407C	23 3/4	5 13/16	4 5/8	7 1/8	2 15/16	3	3 3/8	12 7/8	3	1 1/4
3	7407D	23 3/4	5 13/16	4 5/8	7 1/8	2 15/16	3	3 3/8	12 7/8	3	1 1/4

Specifications

Model	Rated capacity* (tons)	Product code	Reeving	Full load hoist speed** (F.P.M.)	Full load lowering speed** (F.P.M.)	Approx. net weight (lbs.)
Standard model	1	7405C	1	13	18	106
	1	7405D	1	30	42	108
	2	7406C	2	6	9	116
	2	7406D	2	15	21	118
	3	7407C	3	4	6	164
	3	7407D	3	10	14	166
Spark resistant model	1/2	7408B	1	16	22	110
	1	7409B	2	8	11	120
	2	7410B	3	5	7	170

**Hoist speeds based on 90 PSI line pressure with full load at full throttle.

AirStar 6 with hook suspension-Specifications



MODEL TMM-140

Clearance dimensions (in.)

Rated capacity* (tons)	Product code	A	B	C	D	E
1/2	7408A	16 3/4	11 1/16	6 3/4	2 11/16	6 3/16
1	7405A	16 3/4	11 1/16	6 3/4	2 11/16	6 3/16
1	7405B	16 3/4	11 1/16	6 3/4	2 11/16	6 3/16
1	7409A	22 7/8	12 13/16	8 1/16	4 7/16	4 7/8
2	7406A	22 7/8	12 13/16	8 1/16	4 7/16	4 7/8
2	7406B	22 7/8	12 13/16	8 1/16	4 7/16	4 7/8
2	7410A	25 7/8	14 3/8	5 13/16	5 5/8	7 1/8
3	7407A	25 7/8	14 3/8	5 13/16	5 5/8	7 1/8
3	7407B	25 7/8	14 3/8	5 13/16	5 5/8	7 1/8

Specifications

Model	Rated capacity* (tons)	Product code	Reeving	Full load hoist speed** (F.P.M.)	Full load lowering speed** (F.P.M.)	Approximate net weight (lbs.)
Standard model	1	7405A	1	13	18	93
	1	7405B	1	30	42	95
	2	7406A	2	6	9	110
	2	7406B	2	15	21	112
	3	7407A	3	4	6	144
	3	7407B	3	10	14	146
Spark resistant model	1/2	7408A	1	16	22	95
	1	7409A	2	8	11	112
	2	7410A	3	5	7	146

**Hoist speeds based on 90 PSI line pressure with full load at full throttle.

TMM-140 Air Manipulator Hoist

300 Lb./140Kg Air Manipulator Hoist

The CM TMM-140 is ideal for workstation applications. Lightweight and portable, the unit weighs less than 20 pounds and operates at 100% (A5) duty cycle.

- **Smooth operator** - Efficient rotary vane motor provides high uniform torque for quick response, and smooth, quiet operation.
- **Easy to use** - Handle-mounted pendant control for one-handed operation and load positioning.
- **Variable speeds** - Precise and infinite speed control for accurate load spotting.
- **Wire rope design** - Self-winding lifting cable stores conveniently inside unit.
- **Suspension** - Swivel top and bottom hooks with hook latches standard, for portability and quick installation.
- **Disc brake** - Secures load and eliminates drift when pendant is released.
- **Note** - A 3/8" NPT air filter-regulator-lubricator is needed for effective operation. Spark resistant features are not available.

Ideal Applications Include:

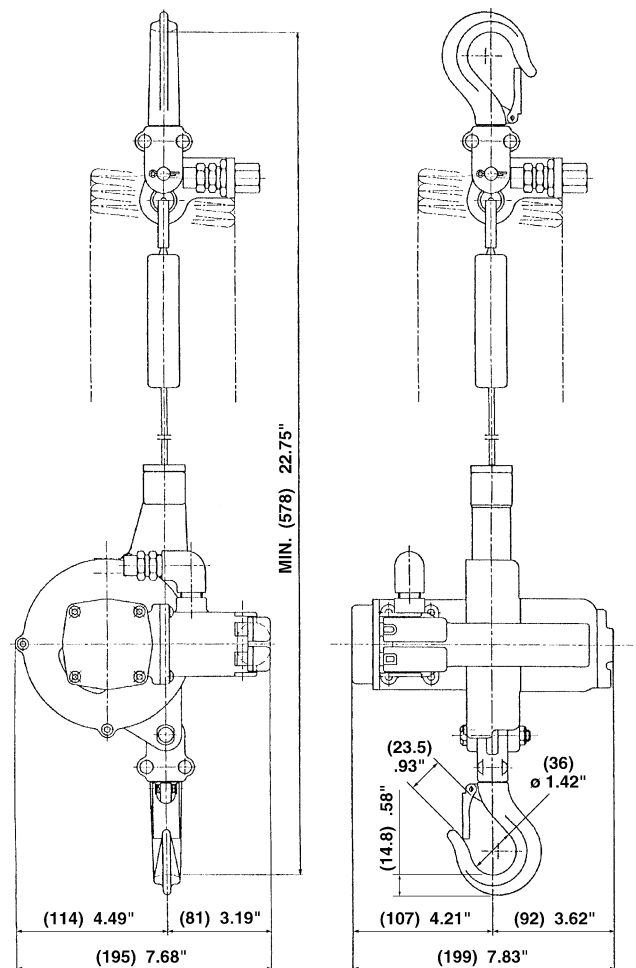
- Assembly lines
- Machine shops
- Cleaning and/or dipping operations
- Manufacturing plants
- Power generating stations
- Shipping departments
- Chemical facilities
- Fabricating shops
- Repetitive lifting applications
- Auto and truck repair
- Handling fixtures and dies

***WARNING**

Overloading and improper use can result in injury.

To avoid injury:

- Do not exceed working load limit, load rating capacity
- Do not use to lift people or loads over people.
- Use only alloy chain for overhead lifting.
- Read and follow all instructions.



Specifications:

Capacity (lb.)	Model number	Product code	Std. lift (ft.)	Lift speeds (FPM)				Air consumption		Headroom (in.)	Housing dimen. (in.)			Max air pres. (psi)	Net wt. (lb.)
				Load (up)	(Down)	No load (up)	(Down)	(m ³ /min)	(ft ³ /min)		D	W	L		
300	TMM-140	17050W	6' 6"	36	59	49	39	0.6	2 1/2	22 3/4	7 2/3	7 7/8	23 3/4	70-95	18.7

NOTE: Dimensions in parenthesis are in millimeters, other dimensions in inches. Top and bottom hook dimensions are the same.